Chapter 14. Permissive Hypercapnia

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Permissive Hypercapnia: Introduction

Carbon dioxide (CO_2) is the "waste product" of aerobic respiration. In health, arterial carbon dioxide tension (Pa_{CO_2}) is tightly regulated, with minute ventilation potently enhanced in response to small elevations in CO_2 tension. In critical illness its role is becoming better understood; indeed, in survivors of cardiac arrest, elevated CO_2 is associated with a higher incidence of reported near-death experiences.[1] Although usually well tolerated, hypercapnia traditionally has been considered to be an adverse event. In fact, the extent and severity of acidosis are predictive of adverse outcome in diverse clinical contexts, including cardiac arrest,[2,3] sepsis,[4–6] and in neonatal practice.[7] Traditional approaches to CO_2 management in the operating room and for patients with acute respiratory failure generally focused on the deleterious effects of hypercapnia, traditionally targeting therefore normocapnia or even hypocapnia.

This approach, however, has been increasingly questioned. The potential for high tidal volumes to injure the lung directly, a phenomenon termed *ventilator-induced lung injury,* is clear from experimental[8,9] and clinical[10–14] studies. Current "protective" ventilator strategies mandate lower tidal volumes (V_T), and generally necessitate hypoventilation and tolerance of hypercapnia. This "permissive hypercapnia" has been accepted progressively in critical care for adult, pediatric, and neonatal patients requiring mechanical ventilation.

Rationale

The potential for mechanical ventilation to contribute to lung and systemic organ injury and to worsen outcome in patients with the acute respiratory distress syndrome (ARDS) is clear. The use of high V_T may cause injury via several mechanisms.[8,9] Increased mechanical stress may activate the cellular and humoral immune response directly in the lung.[8,15–17] Intrapulmonary mediators and pathogens, such as prostaglandins,[18] cytokines,[19] endotoxins,[20] and bacteria,[21] have been demonstrated to access the systemic circulation following high-stretch mechanical ventilation. The demonstration that mechanical ventilation may cause systemic organ dysfunction in animal models could explain in part the high rate of multiorgan failure in ARDS.[22] In current practice, hypercapnia is tolerated as the lesser of two evils so as to realize the benefits of lower tidal volumes.[23,24]

Basic Principles

Permissive Hypercapnia

Conventionally, the protective effect of ventilator strategies incorporating permissive hypercapnia is solely secondary to lower tidal volume, with hypercapnia being permitted so as to achieve this goal. Protective ventilator strategies that involve hypoventilation cause both limitation of lung stretch and elevation of arterial P_{CO_2}; thus, low tidal volume is distinct from elevated P_{CO_2} and, by manipulation of respiratory variables (frequency, V_T, dead space, and inspired CO_2), can to some extent be controlled separately.

Therapeutic Hypercapnia

If hypercapnia were proven to have independent benefit, then deliberately elevating Pa_{CO_2}, termed *therapeutic hypercapnia,* might provide an additional advantage over reducing V_T alone. Conversely, in patients managed with conventional permissive hypercapnia, adverse effects of elevated Pa_{CO_2} may be concealed by the benefits of lessened lung stretch. These issues are underscored by the fact that hypercapnia has adverse effects in some clinical settings, such as critically elevated intracranial pressure or pulmonary vascular resistance. Because patient outcome from critical illness may be related to systemic rather than lung injury, the clinician also must consider the effects on the brain and cardiovascular systems.

These issues have led several investigators to study the direct effects of induced hypercapnia per se in models of lung and systemic organ injury. These studies raise the potential that hypercapnia may exert active roles—beneficial or adverse—in the pathogenesis of inflammation and tissue injury. Thus, therapeutic hypercapnia may someday become a testable clinical approach to critically ill patients.[25]

Accidental Hypercapnia

This is reported most commonly in the context of errors in mechanical ventilation, such as circuit disconnects (or malconnections that permit rebreathing of exhaled gases).[26] A second cause is hypoventilation secondary to drug-induced respiratory depression,[27] severe respiratory failure (e.g., status asthmaticus),[28] or massive aspiration (e.g., grain inhalation).[29]

Determinants of Hypercapnia

The determinants of hypercapnia can be described as the balance of CO_2 production versus elimination.

$$Pa_{CO_2} \ \alpha \ \frac{CO_2 \text{ production}}{CO_2 \text{ elimination}} + \text{inspired } [CO_2] \qquad (1)$$

Inspired CO_2 (FI_{O_2}) usually is negligible. Increased CO_2 production is a potential contributor to hypercapnia, particularly in hypermetabolic disease states such as hyperpyrexia. Therefore, for practical purposes Pa_{CO_2}, reflects the rate of elimination of CO_2; there are no common “physiologic” causes of hypercapnia. Hypercapnia is seen most commonly in the context of acute or chronic respiratory failure associated with reduced minute ventilation or increased dead space.

Effects on Alveolar Gas Exchange

The effects of hypercapnia on systemic oxygenation are discussed below. It is often forgotten that decreasing V_T can reduce oxygenation of pulmonary venous blood. This has two contributing elements: the composition of the alveolar gas and the increase in intrapulmonary shunt. Decreased V_T increases propensity to atelectasis, which increases intrapulmonary shunt.[30] In addition, when V_T is decreased, there is a proportionate decrease in alveolar ventilation ($\dot{V}_A$), resulting in an elevated alveolar CO_2 tension (Pa_{CO_2}). As pointed out by Bidani et al,[31] the alveolar gas equation

$$PA_{O_2} = FI_{O_2}(P_B - 47) - (Pa_{CO_2}/R) \qquad (2)$$

where PA_{O_2} is the alveolar oxygen tension, P_B is the barometric pressure, and R is the respiratory quotient, can be combined with the following relationship:

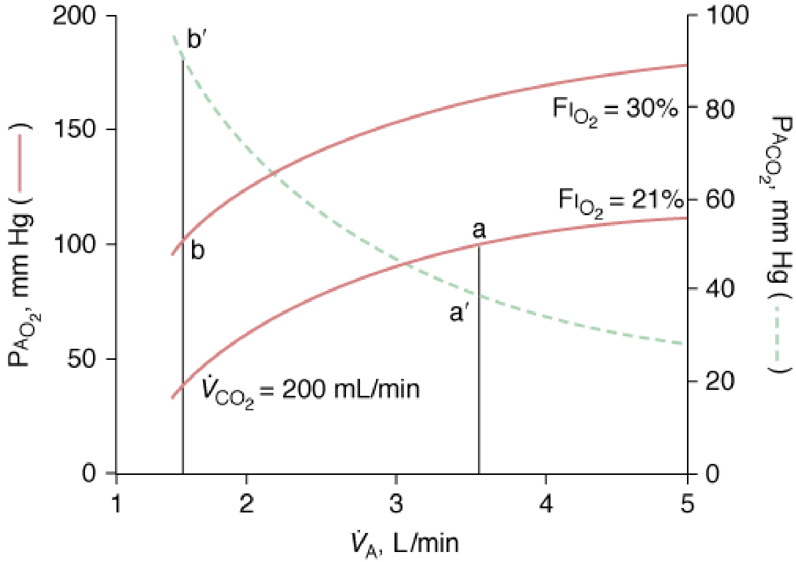
$$Pa_{CO_2} = (\dot{V}_{CO_2}/\dot{V}_A) \times P_B \qquad (3)$$

where $\dot{V}_{CO_2}$ is CO_2 production, to express alveolar O_2 as a function of FI_{O_2} and $\dot{V}_A$:

$$PA_{O_2} = FI_{O_2}(P_B - 47) - (\dot{V}_{CO_2}/\dot{V}_A) \times (P_B/R) \qquad (4)$$

Thus, with the exception of very low levels of minute ventilation, increases in $\dot{V}_A$ have minor impact on alveolar—and consequently pulmonary venous—oxygenation. In contrast, modest increases in FI_{O_2} can have a far more significant impact on oxygenation and can compensate easily for reduced VA (Fig. 14-1).[31]

Figure 14-1



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Relationship of alveolar oxygen tension (PA_{O_2}) and alveolar ventilation ($\dot{V}_A$) at FI_{O_2} (0.21 and 0.30), calculated assuming a constant carbon dioxide production ($\dot{V}_{CO_2}$ of 200 mL/min). At either FI_{O_2}, decreasing $\dot{V}_A$ to low values significantly reduces PA_{O_2}, especially with very low levels of $\dot{V}_A$. A small increase in FI_{O_2} (21% to 30%) easily compensates for the fall in PA_{O_2} resulting from hypoventilation. *Solid line,* relationship between PA_{O_2} and $\dot{V}_A$; *dotted line,* relationship between Pa_{CO_2} and $\dot{V}_A$. (Modified, from Bidani A, Tzouanakis AE, Cardenas VJ Jr, Zwischenberger JB. Permissive hypercapnia in acute respiratory failure. *JAMA.* 1994;272:957–962.)

Physiologic Effects

The physiologic effects of hypercapnia are complex and incompletely understood, with direct effects often counterbalanced by indirect effects. To understand hypercaponia in critically ill patients, the effects of pH and Pa_{CO_2} need to be considered together.

Carbon Dioxide versus Hydrogen Ion

Hypercapnia generally results in acidosis (greater H^+ concentration) via its spontaneous and carbonic anhydrase–catalyzed combination with water to form carbonic acid. The protons thus generated can react with titratable groups in certain amino acids, resulting in structural changes in many proteins and enzymes in cell membranes and cellular aqueous environments.[32] Because acidosis suppresses most cellular functions, the body uses a number of strategies to defend its intracellular and extracellular pH within remarkably narrow limits.[25] The intracellular acidosis produced by hypercapnia may be corrected within a few hours, as opposed to 1 to 2 days for renal compensation.[33] This buffering occurs via active cell-membrane ion transporters that extrude protons and exchange them for extracellular sodium.

During hypercapnia, CO_2 per se may react directly with some free amine groups in proteins to form carbamate residues.[34–36] This binding of CO_2 also modifies protein structure and function and may explain some of the differences in the observed effects of CO_2 and H^+ when both lead to equal changes in pH. A good example is the Bohr effect, where increased Pa_{CO_2} results in a rightward shift of the hemoglobin (Hb)-O_2 dissociation curve, reflecting a lowered affinity of hemoglobin for O_2.

Effects of Hypercapnia on the Lung

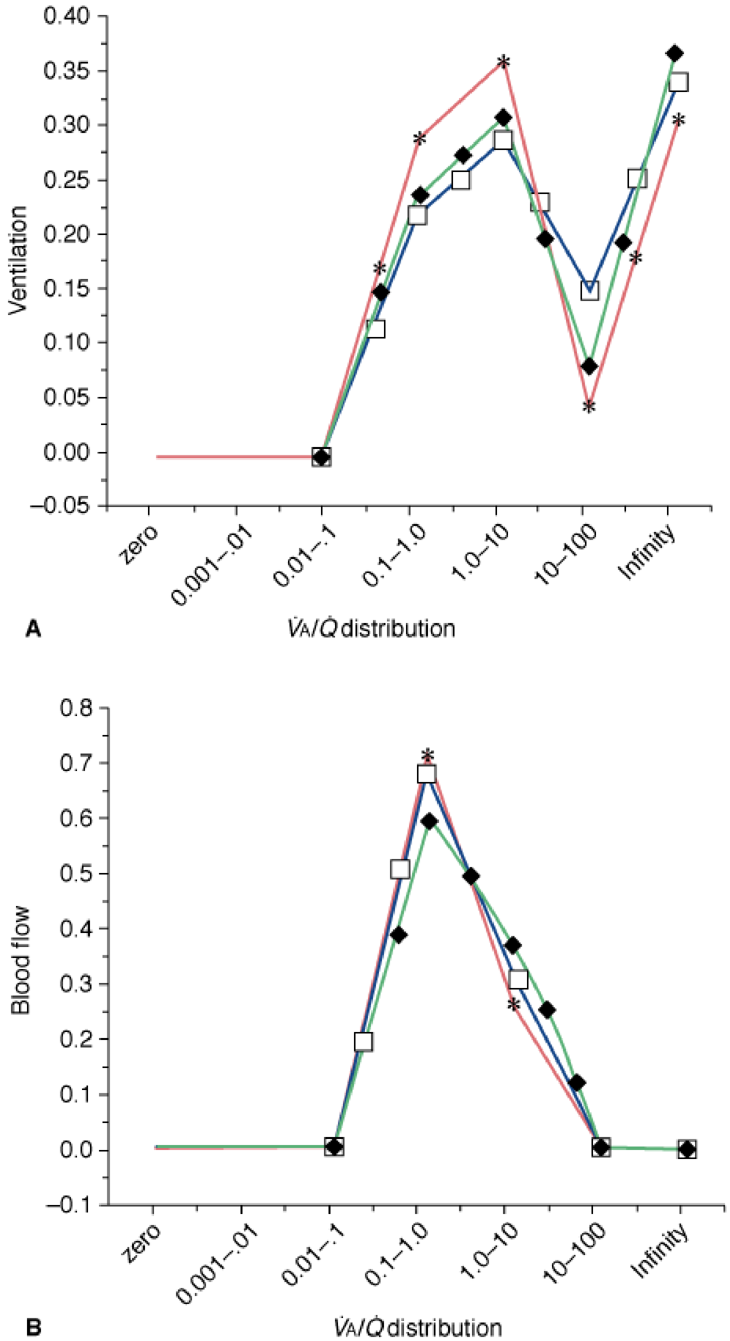
Pulmonary Vascular Effects

In contrast to the systemic circulation, hypercapnic acidosis produces vasoconstriction and increased resistance in the pulmonary circulation;[37] such effects are exacerbated in the setting of preexisting pulmonary hypertension.[37] Hypercapnic vasoconstriction generally is weaker than hypoxic pulmonary vasoconstriction; its more important effect may be in augmenting hypoxic vasoconstriction.[32] Little is understood about how CO_2 acts on pulmonary vascular smooth muscle.

Pulmonary Gas Exchange

Acute respiratory acidosis can alter shunt via autonomic or direct effects on pulmonary vasculature and on the airways. Acidosis enhances hypoxic pulmonary vasoconstriction and therefore usually reduces shunt and increases partial pressure of arterial oxygen (Pa_{O_2}), whereas alkalosis has the opposite effect.[38] CO_2 administration improves matching of ventilation and perfusion and increases arterial oxygenation by this mechanism in both health[39–41] and disease.[42] A dose–response relationship exists wherein increased FI_{CO_2} results in progressive augmentation of Pa_{O_2}.[40,43] In fact, administration of CO_2 during late inspiration—limiting its exposure to the conducting airways—results in most of the beneficial pulmonary effects (i.e., $\dot{V}_A/\dot{Q}$ matching and oxygenation) and less systemic acidosis (Fig. 14-2).[44] Permissive hypercapnia in patients with ARDS appears to increase shunt secondary to a reduction in V_T and airway closure rather than from elevated CO_2 levels.[45]

Figure 14-2



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Effects of room air and inspired CO_2 on the distribution of ventilation (**A**) and of perfusion (**B**). The lowest values of $\dot{V}_A/\dot{Q}$ correspond to regions of intrapulmonary shunt and the highest to regions of alveolar dead space. Addition of CO_2 to inspired gas (throughout the respiratory cycle, *; restricted to late inspiration, □) resulted in more homogeneous ventilation and perfusion compared to no added CO_2 (room air ♦). $\dot{V}_A/\dot{Q}$, ratio of ventilation to perfusion. (Reproduced, with permission, from Brogan TV, Robertson HT, Lamm WJ, Souders JE, Swenson ER. Carbon dioxide added late in inspiration reduces ventilation–perfusion heterogeneity without causing respiratory acidosis. *J Appl Physiol.* 2004;96:1894–1898.)

Airway Tone

Hypercapnia has been reported to either increase[46] or decrease[47] airway resistance. These effects may be explained the direct dilation of small airways and the indirect (i.e., vagally mediated) large airway constriction.[32] These opposing but balanced actions may produce little net alteration in airway resistance.[48]

Lung Compliance

Parenchymal lung compliance increases in response to hypercapnic acidosis. This may be secondary to increased surfactant secretion or more effective surface tension–lowering properties under acidic conditions.[49]

Physiologic Role of Hypercapnia in the Lung

In health, CO_2 alters lung compliance as well as pulmonary vascular and airway tone.[32] The combined effect of small airways constriction and decreased compliance explains the phenomenon of hypocapnic bronchoconstriction and pneumoconstriction that occurs following acute regional pulmonary artery occlusions.[50,51] These effects either may alter regional ventilation to keep pace with a primary change in perfusion or may alter regional perfusion to match a primary change in ventilation.[32]

Central Nervous System Effects of Hypercapnia

Neurovascular Regulation

Hypercapnic acidosis causes cerebral vasodilation. The increase in cerebral blood flow and blood volume must be considered carefully in any patient at risk for raised intracranial pressure.[25] The mechanism of cerebral vasodilation depends on the arterial bed and type of artery. Nakahata et al[52] demonstrated that hypercapnic acidosis induced cerebral precapillary arteriolar vasodilation, which depended on acidosis rather than CO_2. They demonstrated that the adenosine triphosphate (ATP)-sensitive potassium channel plays a major role. Others have suggested roles for both ATP-sensitive and calcium-activated potassium channels[53] and the neuronal isoform of nitric oxide synthase.[54]

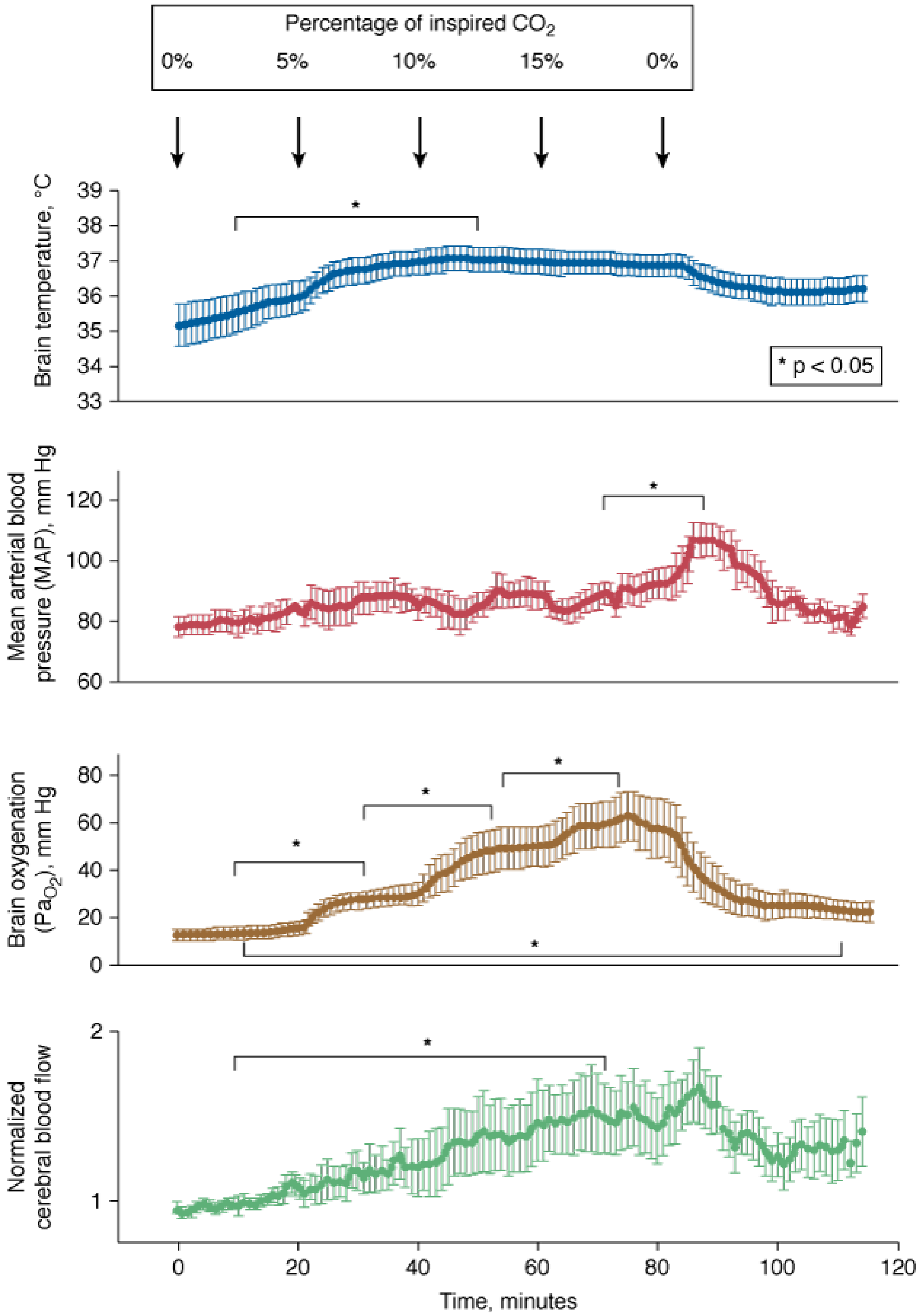
Regulation of Ventilation

Hypercapnia is a potent regulator of ventilation. Mild hypercapnia (increase in end-tidal pressure of carbon dioxide [P_{CO_2}] of 8 mm Hg) in healthy volunteers resulted in a compensatory metabolic alkalosis over 24 to 48 hours that was maintained over the course of exposure.[55] Although there was a modest increase in ventilatory chemosensitivity to acute hypoxia, no change occurred in response to acute elevations in CO_2.

Cerebral Tissue Oxygenation

Hare et al[40] demonstrated that hypercapnic acidosis increases cerebral tissue partial pressure of oxygen (P_{O_2}) through augmentation Pa_{O_2} of and increased cerebral blood flow (Fig. 14-3), although, studies in patients suggest that in the presence of sepsis, hypercapnia may impair cerebral autoregulation.[56] The faster recovery times associated with administered CO_2 in animals[57] and patients[58] following general anesthesia may reflect increased perfusion and anesthetic washout.

Figure 14-3



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Increases in inspired CO_2 concentration produce progressive increases in brain tissue O_2 tension and cerebral perfusion. (Reproduced, with permission, from Hare GM, Kavanagh BP, Mazer CD, et al. Hypercapnia increases cerebral tissue oxygen tension in anesthetized rats. *Can J Anaesth.* 2003;50:1061–1068.)

Neuromuscular Effects of Hypercapnia

Diaphragmatic Function

Recent studies highlight the potential for hypercapnia to exert deleterious neuromuscular effects. Rat diaphragms exposed to prolonged hypercapnia (7.5% CO_2 for 6 weeks) undergo significant changes,[59] including depressed diaphragmatic tension and time to contraction and relaxation, and altered diaphragmatic composition (i.e., increased slow-twitch and decreased fast-twitch fibers). In fact, even short-term exposure (7% inspired CO_2) may impair neuromuscular function transiently through effects on afferent transmission or synaptic integrity in healthy volunteers.[60] Relatively brief exposures to hypercapnia may also cause reversible impairment of diaphragmatic muscle contractility.[61]

Cardiovascular Effects of Hypercapnia

Hemodynamic Effects

Hypercapnic acidosis directly reduces the contractility of cardiac and vascular smooth muscle.[32] This is counterbalanced by the hypercapnic-mediated sympathoadrenal effects causing increased preload, increased heart rate, and decreased afterload, which lead to a net increase in cardiac output.[32] In intact animals and human subjects, myocardial contractility and cardiac output increase during hypercapnia because of increased sympathetic activity.[62]

Effects on Tissue Oxygenation

Hypercapnia results in a complex interaction of altered cardiac output, hypoxic pulmonary vasoconstriction, and intrapulmonary shunt, with a net increase in Pa_{O_2}. Because hypercapnia generally elevates cardiac output, global O_2 delivery is increased.[63] Regional (including mesenteric) blood flow also is increased,[64] thereby increasing organ oxygen delivery. Because hypercapnia and acidosis shift the Hb-O_2 dissociation curve rightward and may cause an elevation in hematocrit,[65] tissue oxygen delivery is further facilitated. Acidosis may reduce cellular respiration and oxygen consumption,[66] which further correct supply–demand imbalance, in addition to enhancing O_2 delivery. Acidosis may protect against ongoing tissue production of *further* organic acids by a negative-feedback loop, providing a mechanism of cellular metabolic shutdown at times of nutrient shortage (e.g., ischemia).[67] In addition, hypercapnic acidosis increases P_{O_2} in subcutaneous tissues and in the intestinal wall.[68]

Hypercapnia and Development

Hypercapnia may exert complex effects on development,[69] such as inhibition of embryonic morphogenesis, as well as egg laying and hatching in the *Drosophila.*[70] It induces aberrant motility and slows development, decreases fertility, and increases life span in *Caenorhabditis elegans.*[71] Tissue effects in neonates may differ from those in adults as exposure to hypercapnia for 2 weeks reduced expression of several matrix proteins in infant, but not in adult, lungs,[72] and sustained exposure in the neonate may cause microvascular injury and impair brain growth via induction of nitrative stress.[73] In the neonatal lung, hypercapnia may increase alveolar budding, but may also increase central nervous system apoptosis.[74]

Cellular and Molecular Effects of Hypercapnia

A clear understanding of cellular and biochemical mechanisms underlying the effects of hypercapnia is essential for successful translation of laboratory findings to the bedside, as well as for prediction of potential side effects.

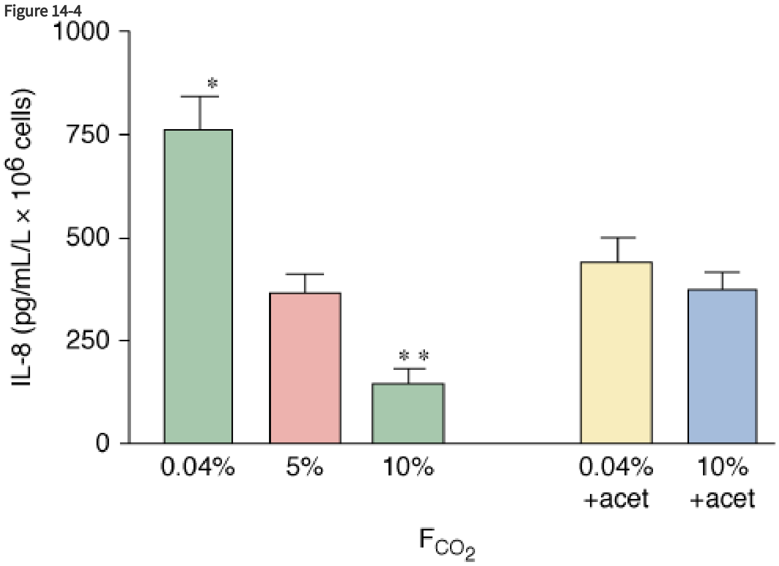
Effects on the Immune Response

Hypercapnia and acidosis modulate diverse components of the host immune response, particularly cytokine and chemokine signaling, as well as neutrophil and macrophage function.

Chemokine and Cytokine Signaling

Hypercapnia interferes with coordination of the immune response by modulating signaling among immune effector cells. Hypercapnic acidosis inhibits neutrophil release of interleukin (IL)-8 following endotoxin stimulation[75] (Fig. 14-4), reduces release of tumor necrosis factor-α (TNF-α) and IL-1 from stimulated macrophages in vitro,[76] and inhibits IL-6 and TNF-α in macrophages[77] as well as in whole (human) blood.[78] Hypercapnic acidosis reduces TNF-α concentrations in the bronchoalveolar fluid following in vivo pulmonary ischemia-reperfusion.[79] Such effects on cytokines and chemokines may be sustained; for example, intraperitoneal macrophages demonstrate impaired TNF-α production for up to 3 days following exposure to hypercapnia.[80,81] These actions appear to be mediated, at least in part by inhibition of the transcriptional regulator nuclear factor kappa B (NF-κB).[82]

Figure 14-4



Hypercapnia (10% CO_2) inhibits, whereas hypocapnia (0.04% CO_2) potentiates, the release of interleukin (IL)-8 from endotoxin-stimulated neutrophils. Incubation with acetazolamide, which impairs the effect of extracellular CO_2 on intracellular pH, abolished the influence of extracellular CO_2. (Reproduced, with permission, from Coakley RJ, Taggart C, Greene C, McElvaney NG, O'Neill SJ. Ambient pCO_2 modulates intracellular pH, intracellular oxidant generation, and interleukin-8 secretion in human neutrophils. *J Leukoc Biol.* 2002;71:603–610.)

The Cellular Immune Response

Hypercapnia may impact the cellular immune response via several mechanisms, including impaired coordination of cytokine signaling as well as inhibition of neutrophil expression of key inflammatory and adhesion molecules (e.g., chemokines, selectins, and intercellular adhesion molecules).[75] Neutrophil chemotaxis may be directly impaired.[83] These effects also occur in vivo as lung neutrophil recruitment is inhibited during ventilator-induced[84] and endotoxin-induced[85] lung injury. Hypercapnic acidosis directly impairs neutrophil[86] and macrophage[70] phagocytosis, effects that appear to be a function of the acidosis rather than the CO_2 tension.[87]

The cellular and molecular mechanisms underlying the inhibitory effects of hypercapnic acidosis in the neutrophil and macrophage are increasingly understood. Both hypercapnia and acidosis impair neutrophil intracellular pH regulation. Intracellular pH decreases when neutrophils are activated by immune stimuli[75] and, where pH is normal, there tends to be a recovery in neutrophil intracellular pH (toward normal). Hypercapnia decreases extracellular and intracellular pH in the local milieu, resulting in a rapid fall in neutrophil cytosolic pH,[88] potentially overwhelming the capacity of phagocytes—especially when activated[89]—to regulate cytosolic pH.

Free Radical Generation and Activity

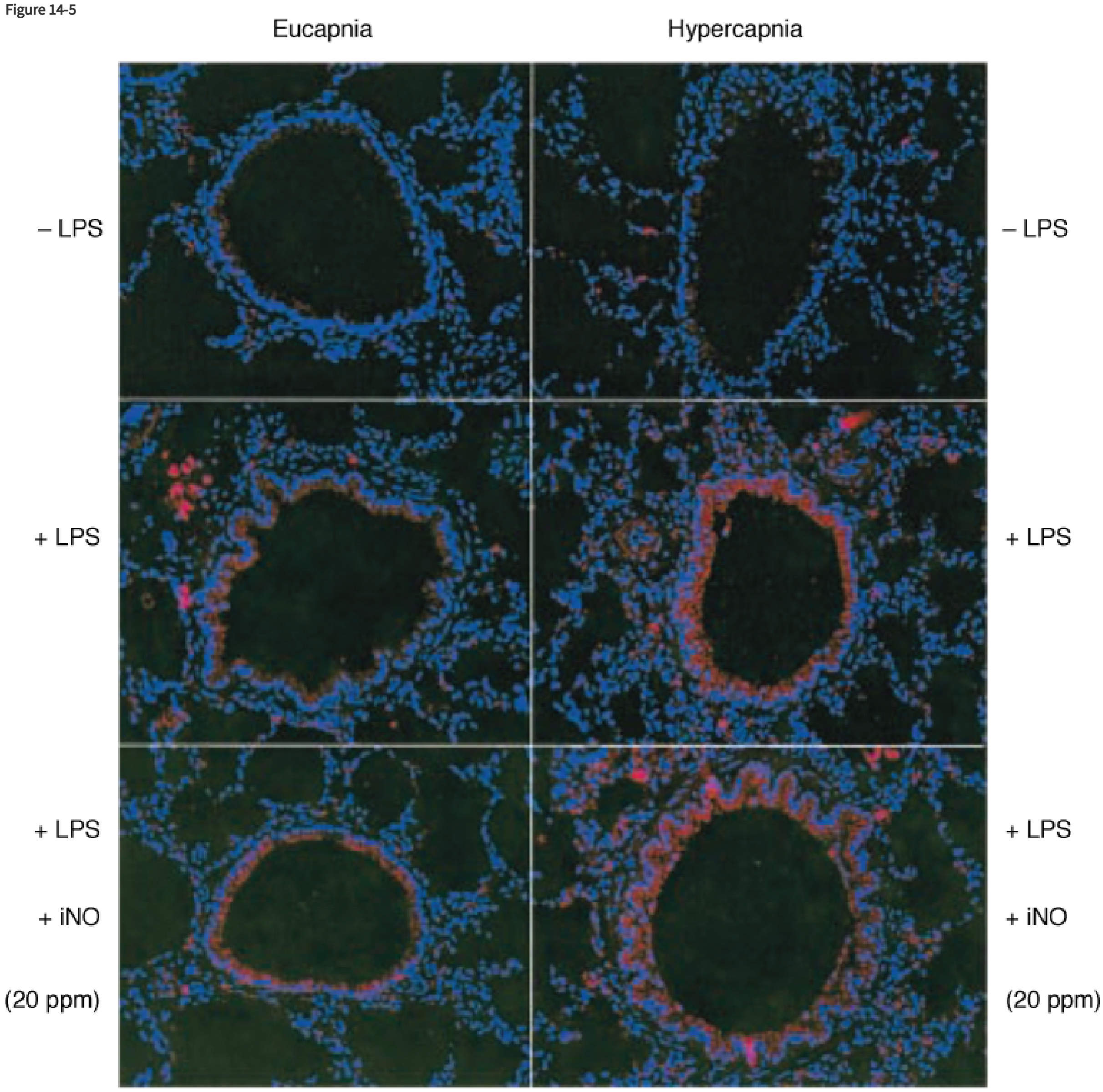
In common with many biologic systems, the enzymes that produce oxidizing free radicals function optimally at physiologic pH. Oxidant generation by both basal and stimulated neutrophils appears to be regulated by ambient CO_2 levels, with oxidant generation reduced by hypercapnia and increased by hypocapnia.[75] Production of superoxide by stimulated neutrophils in vitro is decreased at acidic pH.[90] Hypercapnic acidosis inhibits the generation of oxidants such as superoxide by neutrophils[75] and by macrophages.[91] Although free radicals such as superoxide have been implicated in the pathogenesis of injury in acute lung injury and/or ARDS, effective phagocyte generation of free radicals is necessary for killing of ingested bacteria. The finding that hypercapnia reduces endotoxin-induced pulmonary oxidant production[92] reinforces concerns that neutrophil-mediated bacterial killing might be critically inhibited.

In the brain, hypercapnic acidosis attenuates glutathione depletion and lipid peroxidation,[93] which reflect free-radical activity and tissue damage, respectively. In the lung, hypercapnic acidosis reduces free-radical tissue injury following ischemia–reperfusion[79] and attenuates the production of the higher oxides of nitric oxide, such as nitrate and nitrite, following both ventilator-induced[94] and endotoxin-induced[85] injury. Hypercapnic acidosis inhibits injury mediated by xanthine oxidase and directly inhibits the enzyme.[95] Interactions between CO_2 and free radicals may also have fundamental effects as recent work indicates that hypercapnia may cause signaling, in part, by cellular oxidation.[96]

Peroxynitrite-Mediated Tissue Nitration

Specific concerns exist regarding the potential for hypercapnia to potentiate tissue nitration by peroxynitrite, a potent free radical. Peroxynitrite is produced in vivo largely by the reaction of nitric oxide with superoxide radical and causes tissue damage by oxidizing a variety of biomolecules and by nitrating phenolic amino acid residues in proteins.[97] The potential for buffered hypercapnia to promote the formation of nitration products from peroxynitrite has been clearly demonstrated in vitro.[98,99] The potential, however, for hypercapnic acidosis to promote nitration of lung tissue in vivo depends on the injury. For example, hypercapnic acidosis decreased tissue nitration following pulmonary ischemia–reperfusion,[79] but increased nitration following endotoxin exposure[85,98,100] (Fig. 14-5).

Figure 14-5



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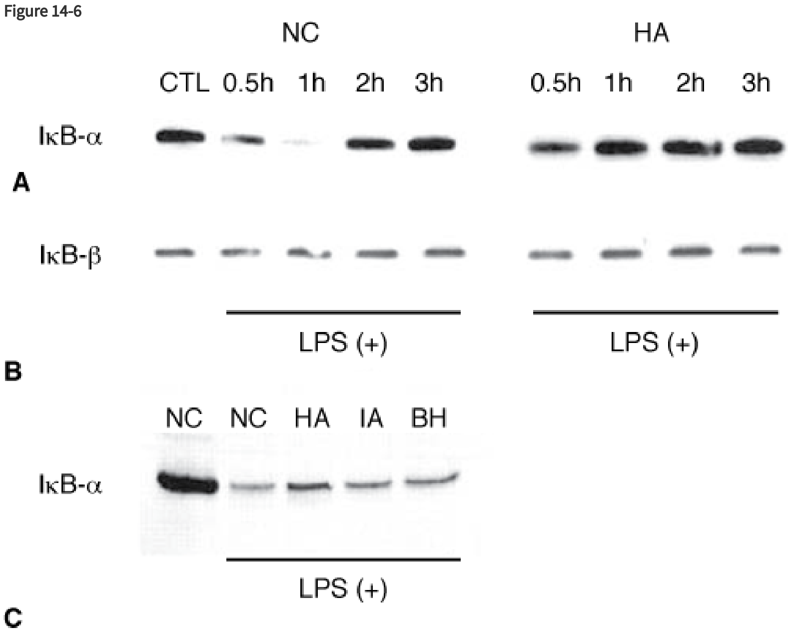
Hypercapnia and inhaled nitric oxide significantly increased the formation of 3-nitrotyrosine following lipopolysaccharide pretreatment. *HC,* Hypercapnia; *iNO,* inhaled nitric oxide; *LPS,* lipopolysaccharide. (Reproduced, with permission, from Lang JD, Figueroa M, Sanders KD, Aslan M, Liu Y, Chumley P, Freeman BA. Hypercapnia via reduced rate and tidal volume contributes to lipopolysaccharide-induced lung injury. *Am J Respir Crit Care Med.* 2005;171:147–157.)

Regulation of Gene Expression

Some effects of hypercapnia may be mediated by regulation of gene expression. Several unidentified proteins are upregulated by hypercapnia in the normal lung,[101] and hypercapnia may regulate this process at the level of gene transcription via altering the half-life of messenger RNA or by modulating protein synthesis. Molecular mechanisms underlying hypercapnic acidosis–mediated control of gene transcription include membrane acid-sensing ion channels and acid-responsive gene promoter regions.[102–104] Furthermore, the coding for certain proteins for messenger RNA has pH-sensitive regions.[103]

Hypercapnic acidosis has been demonstrated to regulate the expression of genes central to the inflammatory response in models of cell injury. NF-κB is a key regulator of the expression of multiple genes involved in inflammatory response, and its activation represents a pivotal early step in activation of the inflammatory response.[105] NF-κB is found in the cytoplasm in an inactive form bound to inhibitory proteins called *inhibitory protein* κB (IκB). The important isoforms are IκB-α and IκB-β. IκB proteins are phosphorylated by the IκB kinase complex and subsequently degraded, thus allowing NF-κB to translocate into the nucleus, bind to specific promoter sites, and activate target genes.[105] Hypercapnic acidosis inhibits endotoxin-induced NF-κB activation and DNA binding in pulmonary endothelial cells by decreasing IκB-α degradation[82] (Fig. 14-6). Hypercapnic acidosis also suppressed endothelial production of intercellular adhesion molecule-1 and IL-8, which are critically regulated by the NF-κB pathway.[82] Importantly, although inhibition of NF-κB may have important antiinflammatory effects, it may also reduce repair following injury.[106]

Figure 14-6



Hypercapnia suppresses the degradation of IκB-α **(A)** but not IκB-β **(B)** following exposure to lipopolysaccharide, thereby inhibiting the nuclear translocation of NF-κB and downstream cytokine production. The effects of isocapnic acidosis and buffered hypercapnia **(C)** on IκB-α degradation were intermediate between normocapnic control and hypercapnic acidosis conditions. *BH,* buffered hypercapnia; *HA,* hypercapnic acidosis; *IA,* isocapnic acidosis; *LPS,* lipopolysaccharide; *NC,* normocapnia; *NF-κB,* nuclear factor κB. (Reproduced, with permission, from Takeshita K, Suzuki Y, Nishio K, et al. Hypercapnic acidosis attenuates endotoxin-induced nuclear factor-[kappa]B activation. *Am J Respir Cell Mol Biol.* 2003;29:124–132.)

Effects on Recovery and Repair Mechanisms

Extracellular Fluid Clearance

Hypercapnia appears to inhibit epithelial fluid transport, a potentially important mechanism by which alveolar edema is cleared during lung injury.[107] The effect appears to be mediated via hypercapnia per se and seems to be independent of ambient pH.[107] The mechanism of action involves adenosine monophosphate (AMP)-activated protein kinase, which, when activated by hypercapnia, promotes Na, K-ATPase endocytosis.[108] The decrement in alveolar fluid clearance can be prevented by β-adrenergic agonists or cyclic adenosine monophosphate.[108] More recent studies have also implicated extracellular signal-regulated kinase in the activation of AMP-activated protein kinase by hypercapnia, leading to Na, K-ATPase endocytosis.[109]

Epithelial Wound Repair

Hyercapnia appears to inhibit pulmonary epithelial-cell resealing[110] following ventilator-induced lung injury via a pH-dependent mechanism,[111] and also inhibits the closure of pulmonary epithelial wounds via a mechanism that involves reduced activation of NF-κB, which, in turn, inhibits cellular migration.[106] These effects of hypercapnia on wound healing appear to be a function of CO_2 tension rather than acidosis.

Effects in the Setting of Organ Injury

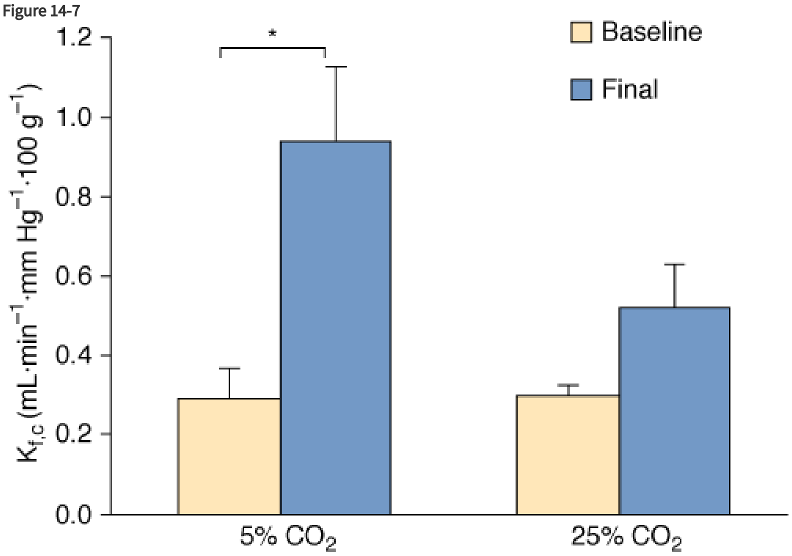
Hypercapnia and the Injured Lung

Insights from Laboratory Data

Ischemia–Reperfusion Injury

Hypercapnic acidosis attenuates the increased lung permeability consequent to free-radical-mediated lung injury (Fig. 14-7).[95] Although hypercapnic acidosis reduces xanthine oxidase activity,[95] this does not account for all its protective effects.[112] Subsequent in vivo studies confirmed and further characterized the protective effects of hypercapnic acidosis in ischemia–reperfusion lung injury. Hypercapnic acidosis preserved lung mechanics, attenuated protein leakage and reduced pulmonary edema in addition to preserving oxygenation in comparison with control conditions following in vivo pulmonary ischemia–reperfusion,[79] as well as in lung injury secondary to splanchnic reperfusion.[43] Such protective effects of hypercapnic acidosis are not mediated via decreases in pulmonary artery resistance; on the contrary, protection occurred despite elevated pulmonary artery pressures.[43]

Figure 14-7



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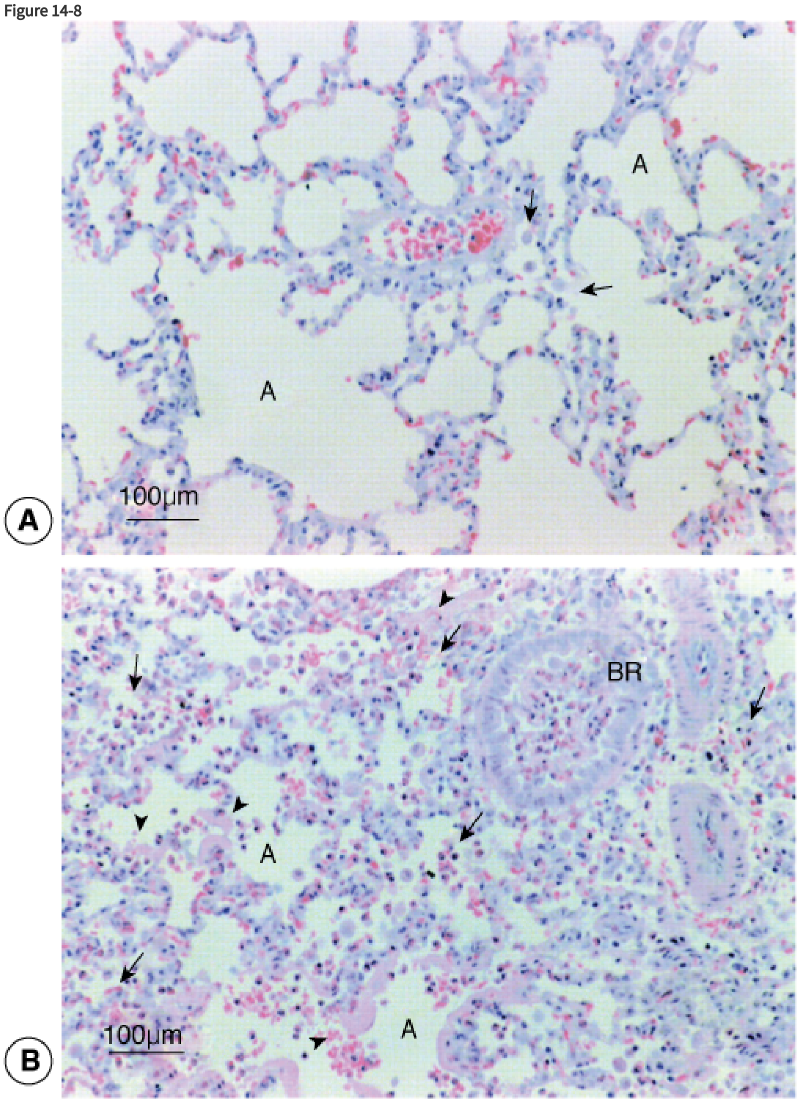
Effects of CO_2 on ischemia–reperfusion in the isolated perfused lung. Pulmonary microvascular permeability is significantly less following reperfusion in the presence of hypercapnia (25% CO_2) compared with normocapnia (5% CO_2). $K_{f,c}$, pulmonary microvascular filtration coefficient. (Reproduced, with permission, from Shibata K, Cregg N, Engelberts D, et al. Hypercapnic acidosis may attenuate acute lung injury by inhibition of endogenous xanthine oxidase. *Am J Respir Crit Care Med.* 1998;158(5 Pt 1):1578–1584.)

Ventilation-Induced Lung Injury

The direct effects of hypercapnic acidosis in ventilator-induced lung injury have been examined in both ex vivo and in vivo models. The addition of inspired CO_2 decreased ventilator-induced lung injury in the isolated rabbit lung,[94] the in vivo rabbit[84] (Fig. 14-8), and in the in vivo mouse.[113] Not all the data are positive. Supplemental CO_2 exhibits more modest protective effects in the setting of more clinically relevant tidal stretch. Strand et al[114] demonstrated that significant hypercapnic acidosis (mean Pa_{CO_2} of 95 mm Hg) was well tolerated in preterm lambs and also appeared to reduce lung injury. In the context of a clinically relevant high V_T adult model (V_T 12 mL/kg, positive end-expiratory pressure 0 cm H_2O), Laffey et al[43] reported that hypocapnia was deleterious and conversely that hypercapnic acidosis was protective. In contrast, inspired CO_2 did not attenuate lung injury in an atelectasis-prone model that mimics neonatal respiratory distress syndrome.[115] Taken together, these findings suggest that while hypercapnic acidosis substantially attenuates injury secondary to excessive stretch, its effects in the context of more clinically relevant lung stretch or, with extensive atelectasis, may be more modest.

Figure 14-8



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Lung histology following high tidal volume ventilation in the in vivo rabbit. The extent of histologic injury is far less with addition of inspired CO_2 (Pa_{CO_2} of 80 to 100 mm Hg) **(A)** than with a Pa_{CO_2} of 40 mm Hg **(B).** (Sinclair SE, Kregenow DA, Lamm WJ, Starr IR, Chi EY, & Hlastala MP. Hypercapnic acidosis is protective in an in vivo model of ventilator-induced lung injury. *Am J Respir Crit Care Med*. 2002;166:403–408.)

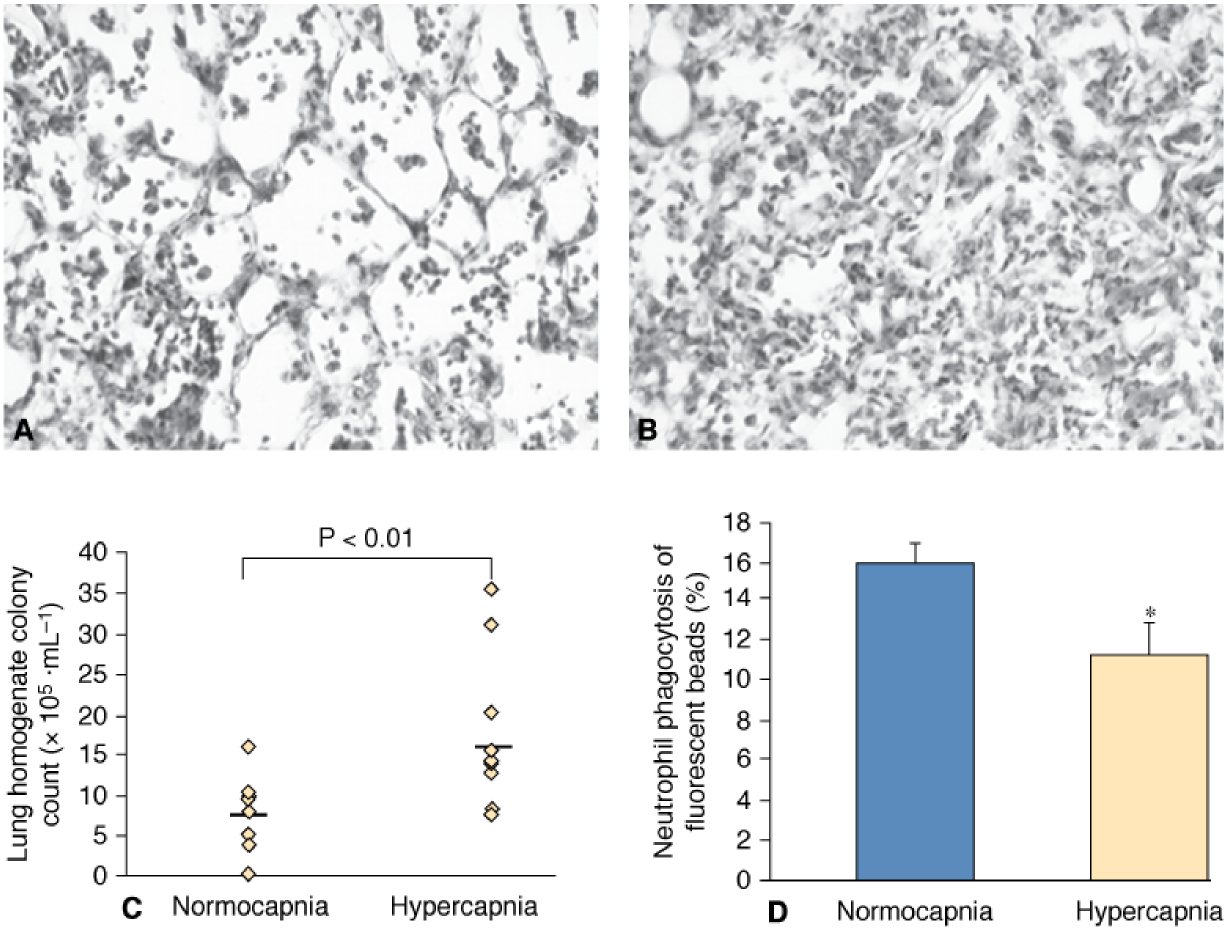
Pulmonary Hypertension

In neonatal models, chronic hypercapnia attenuates the development of hypoxia-induced pulmonary hypertension[116,117] in part via augmentation of L-arginase expression resulting in increased local nitric oxide[117] and in part via inhibition of endothelin expression.[118] Although no mechanism was established, comparable effects were described wherein chronic hypercapnia attenuated the development of long-term hypoxia-induced pulmonary hypertension in the rat.[119]

Pulmonary Sepsis

The mechanisms of lung injury in sepsis-induced ARDS are quite distinct from those in “sterile” experimental models. Lipopolysaccharide, a key endotoxin of gram-negative bacteria, initiates lung injury by activating a specific receptor, the Toll-like receptor-4.[120] Hypercapnic acidosis can reduce the severity of lung injury induced by local administration of endotoxin.[85] Mechanisms of sepsis-induced injury, however, include damage to host tissue from the inflammatory response, as well as damage caused by direct bacterial invasion.[24,85] In preclinical sepsis models, the effects of hypercapnia appear to depend also on the severity—and phase—of the injury. Although hypercapnic acidosis did not alter the severity of mild injury induced by instilled *Escherichia coli,*[121] it did reduce the severity of more severe lung injury induced by the same organism.[122] In the setting of established *E. coli* pneumonia, short-term hypercapnia reduced lung injury, particularly when effective antimicrobial therapy was used.[122] In contrast, in the setting of prolonged *E. coli*-induced pneumonia (i.e., 72 hours), ongoing environmental hypercapnia worsened the severity of the lung injury.[86] Here, prolonged hypercapnia reduced bacterial killing, in part by reduced neutrophil phagocytic activity (Fig. 14-9).[86] Subsequent studies confirmed that these effects in prolonged infection appear to be a function of the hypercapnia and not acidosis, as buffered hypercapnia also worsened infection-induced injury.[87] Nonetheless, deleterious effects of hypercapnia in the setting of prolonged pneumonia were completely ablated by antimicrobial therapy.[86]

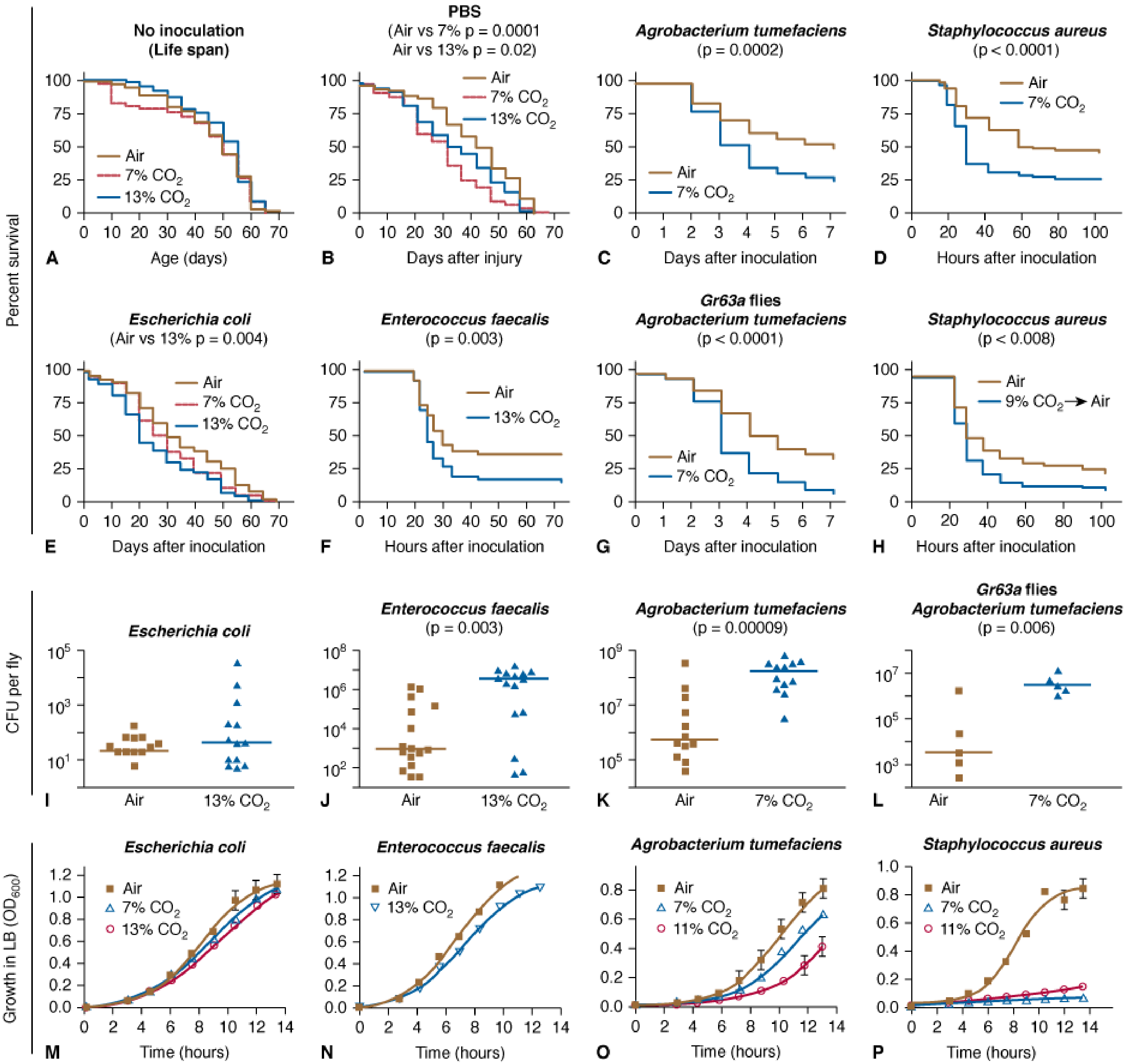
Figure 14-9



Sustained hypercapnic acidosis worsens pneumonia-induced lung injury and increases bacterial load. *Panels A* and *B* represent photomicrographs of section of lung tissue from a lung exposed to normocapnia and hypercapnia respectively, 2 days after intratracheal infection with *E. coli.* Animals exposed to environmental hypercapnia (inspired CO_2 5%) sustained a more severe lung injury. *Panel C* demonstrates greater bacterial load in lungs from *E. coli* infected groups exposed to hypercapnia compared to normocapnia. *Panel D* demonstrates that neutrophils from rats exposed to hypercapnia have a reduced ability to phagocytose fluorescent latex beads compared to neutrophils from normocapnic rats. (Modified, with permission, from O'Croinin DF, Nichol AD, Hopkins N, et al. Sustained hypercapnic acidosis during pulmonary infection increases bacterial load and worsens lung injury. *Crit Care Med.* 2008;36:2128–2135.)

Concerns regarding harm caused by hypercapnia in prolonged lung infection are reflected in an elegant series of studies in the *Drosophilia*[70] demonstrating an increased mortality in infections with multiple different bacterial pathogens, including *Staphylococcus aureus, Enterococcus faecalis,* and *E. coli.*[70] The effects appear to be a function of the elevated CO_2 (not lowered pH) and are mediated in part via suppression of Rel/NF-κB, an important conserved pathway (Fig. 14-10).[70]

Figure 14-10



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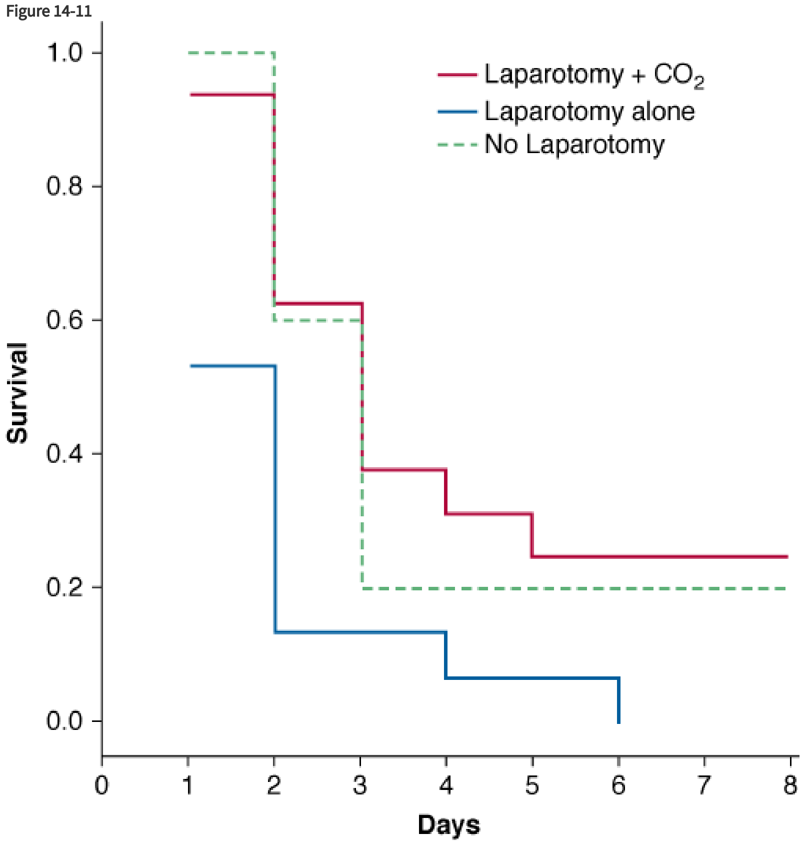
Prolonged hypercapnia decreases resistance of *Drosophila* to specific bacterial infections. *Panel A.* Hypercapnia does not affect *Drosophila* life span. *Panels B* to *H.* Hypercapnia slightly increases death of flies inoculated with sterile phosphate-buffered saline (*PBS*) (*Panel B*) or with *E. coli* (*Panel E*), but significantly increases mortality at CO_2 levels as low as 7% after inoculation with *A. tumefaciens* (*Panel C*), the human pathogen *S. aureus* (*Panel D*), and the *Drosophila* natural pathogen *E. faecalis* (*Panel F*). Immune suppression does not require the neuronal CO_2 receptor Gr63a (*Panel G*). *Panel H.* Pretreatment of flies with 9% CO_2 before *S. aureus* infection in air is sufficient to increase mortality, even when flies are cultured in air after inoculation. For *Panels A* to *H,* unless otherwise noted, flies were exposed to indicated CO_2 level for 24 hours before inoculation and returned to hypercapnia until end of assay. We show representative results for the lowest CO_2 levels at which significant effects on mortality were consistently observed. *Panels I* to *L.* Hypercapnia increases the bacterial load for strains causing increased mortality during hypercapnia. *Horizontal lines* show medians. *Panels M* to *P.* Effects of hypercapnia on bacterial growth. Note that *S. aureus* growth is dramatically reduced in 7% CO_2 even though hypercapnia increases mortality of flies infected with *S. aureus.* (Reproduced, with permission, from Helenius IT, Krupinski T, Turnbull DW, et al. Elevated CO_2 suppresses specific *Drosophila* innate immune responses and resistance to bacterial infection. *Proc Natl Acad Sci U S A.* 2009;106(44):18710–18715.)

Systemic Sepsis

In a sheep model of established sepsis, hypercapnia augmented hemodynamics (comparably to dopamine) and resulted in increased systemic oxygenation.[123] In a small animal model of established systemic sepsis induced by cecal ligation and perforation, hypercapnia had similar effects on hemodynamics.[124] In contrast to established pulmonary sepsis,[86] hypercapnia reduced the lung injury—in terms of histology and lung mechanics—associated with cecal ligation and perforation[124]; importantly, in this setting, hypercapnia did not increase bacterial load in the lung, bloodstream, or peritoneal cavity.[124] Buffering the hypercapnic acidosis maintained the hemodynamic benefits, but ablated the lung protection.[125] Direct intraabdominal administration of carbon dioxide—by introducing operative pneumoperitoneum—may reduce the severity of abdominal sepsis. In experimental endotoxin sepsis, CO_2 pneumoperitoneum reduced mortality following laparotomy,[126] and insufflation before laparotomy also increased survival.[127] These protective effects are seen also in polymicrobial sepsis (Fig. 14-11),[128,129] and may be secondary to the immunomodulatory effects of hypercapnia,[127,130] which appear to be mediated by the localized peritoneal acidosis.[131,132]

Figure 14-11



Insufflation of CO_2 into the peritoneal cavity improves survival following cecal ligation and puncture-induced systemic sepsis. Animals were first subjected to cecal ligation and puncture. Four hours later, animals underwent a laparotomy and induction of a CO_2 pneumoperitoneum (laparotomy + CO_2), laparotomy alone, or no laparotomy, and survival was determined over the following 8 days. (Modified, with permission, from Metzelder M, Kuebler JF, Shimotakahara A, et al. CO_2 pneumoperitoneum increases survival in mice with polymicrobial peritonitis. *Eur J Pediatr Surg.* 2008;18:171–175.)

Insights from Other Models

Inhaled CO_2 protects against hyperoxic lung injury in the neonatal rat.[116] In lung injury induced by oleic acid, metabolic alkalosis increased intrapulmonary shunt.[133] Whereas metabolic acidosis decreased shunt and increased arterial oxygen saturation (SaO_2) and mixed venous oxygen tension (PvO_2), hypercapnic acidosis did not affect intrapulmonary shunt fraction ($\dot{Q}_S/\dot{Q}_T$), Sa_{O_2} or mixed venous oxygen saturation

(SvO_2). Both Pa_{O_2} and PvO_2, however, increased with hypercapnic acidosis possibly in part because of a rightward shift of the Hb-O_2 dissociation curve and/or increased cardiac output. When the hypercapnia was buffered with bicarbonate, gas exchange deteriorated during hypercapnia, shunt increased, and was reduced.[133] The direct vasodilator effect of hypercapnia in hypoxic lung regions may be opposed by acidosis with little overall net effect on shunt from respiratory acidosis. With buffering however, the vasodilator effect of CO_2 may be unopposed, hypoxic pulmonary vasoconstriction inhibited, and shunt increased.[133]

Therapeutic versus Permissive Hypercapnia

The effects of hypercapnia may also depend on the means of achieving it. The effects of hypercapnia in preclinical models described earlier were generally obtained by increasing CO_2 in the inspired gas. Hypercapnia, however, can be induced by reducing V_T and/or respiratory rate; achieved this way, hypercapnia appears to worsen lung damage induced by systemic endotoxin administration.[100] Inspired CO_2 is distributed uniformly through all ventilated regions, whereas hypercapnia caused by reducing alveolar ventilation may not be.[134] These issues underline the need to consider the means of achieving hypercapnia, as well as the diversity of experimental models.[135]

Insights from Clinical Data

Reduction of V_T results in permissive hypercapnia. The lung alterations, however, involve the reduced V_T, the ensuing hypercapnic acidosis, and the potential influence of altered respiratory frequency. No clinical studies to date have dissected out these changes; it is difficult to conceive of how this would be done. Nonetheless, conventional application of permissive hypercapnia involves reduction of V_T, some increase in respiratory frequency, and tolerance of hypercapnic acidosis.

Pulmonary Vascular Effects

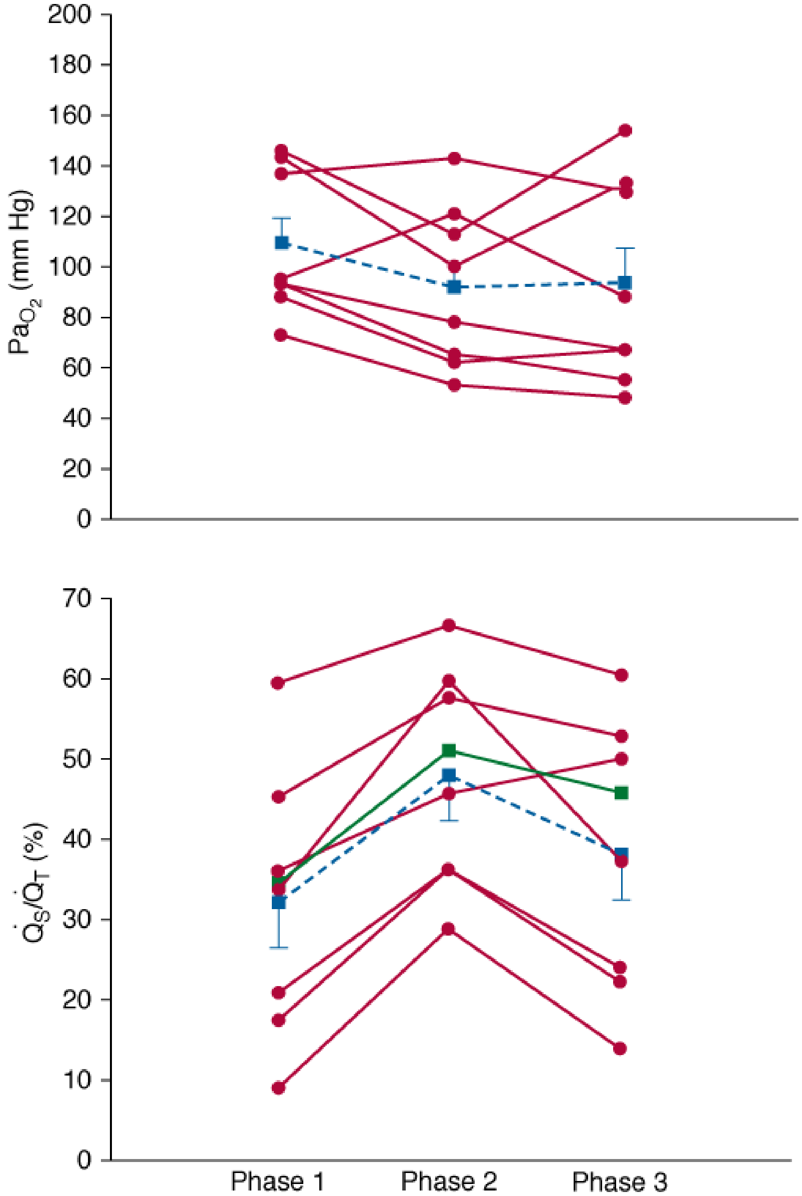
Pulmonary hypertension is almost invariable in ARDS. While protective ventilation may augment pulmonary vascular flow, the net effect of hypercapnic acidosis is usually to increase pulmonary vascular resistance. Inhaled nitric oxide usually can overcome such hypercapnia-induced pulmonary hypertension and may increase cardiac output.[136] Pulmonary hypertension may result in high capillary wall stress; therefore, worsening of such stress by hypercapnia theoretically could exacerbate stretch-induced lung injury.[33,137]

Pulmonary Gas Exchange in Acute Respiratory Distress Syndrome

In ARDS, hypercapnia usually results in a slight increase in Pa_{O_2} and a somewhat larger increase in venous and tissue P_{O_2}.[138] The increase in Pa_{O_2} occurs partly because of an increase in cardiac output and partly because of a rightward shift of the Hb-O_2 dissociation curve, which facilitates oxygen unloading to the tissues.[25,139] The overall effect, therefore, is likely to enhance tissue oxygen uptake. In patients with ARDS, the reduction in mean airway pressure after initiation of pressure-limited ventilation (without high positive end-expiratory pressure [PEEP]) may result in lung derecruitment and increased shunt.[140] Three clinical studies shed light on these mechanisms.

Feihl et al[45] provided a detailed assessment of pulmonary physiology associated with a large reduction in V_T (mean: from 10 to 6 mL/kg). Induction of permissive hypercapnia markedly increased intrapulmonary shunt, although there was no effect on dispersion of $\dot{V}_A/\dot{Q}$ (Fig. 14-12).[45] Permissive hypercapnia increased cardiac output and decreased Pa_{O_2} from 109 to 92 mm Hg apparently owing to a combined effect of reduced V_T, increased shunt fraction, and decreased alveolar ventilation.[45] Thorens et al[138] studied the rapid induction of hypercapnia in eleven patients with ARDS, where V_T was reduced such that Pa_{CO_2} rose from approximately 40 to 60 mm Hg, and pH correspondingly decreased from 7.4 to 7.26.[138] There were significant increases in venous mixture, cardiac index, and mean pulmonary artery pressure, prompting the authors to caution against rapid induction of permissive hypercapnia. Pfeiffer et al[141] studied the effect of permissive hypercapnia consequent to V_T reduction in twenty-two patients with ARDS categorized into septic (i.e., hyperdynamic) or nonseptic groups. Multiple inert gas uptake measurements revealed an increase in intrapulmonary shunt but maintained—or increased—Pa_{O_2}.[141]

Figure 14-12



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The effects of permissive hypercapnia and dobutamine on arterial oxygen tension (Pa_{O_2}) and venous admixture $\dot{Q}_S/\dot{Q}_T$. Phase 1 (high V_T, 10.3 mL/kg) represents baseline conditions. Phase 2 (low V_T, 6.5 mL/kg) represents a change to permissive hypercapnia, which was associated with a slight decrease in mean Pa_{O_2} and a large increase in $\dot{Q}_S/\dot{Q}_T$. Phase 3 represents resumption of baseline high V_T (10.3 mL/kg) plus infusion of dobutamine; it resulted in stabilization of Pa_{O_2} despite residual elevation of intrapulmonary shunt. (Reproduced, with permission, from Feihl F, Eckert P, Brimioulle S, et al. Permissive hypercapnia impairs pulmonary gas exchange in the acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 2000;162:209–215.)

Overall, these data suggest two opposing sets of effects on oxygenation: reduced V_T worsens atelectasis and increases intrapulmonary shunt, but these effects are countered by elevated cardiac output (and perhaps reduced O_2 consumption), which increases mixed venous O_2 content.[141] These concepts have been supported by mathematical models of oxygen kinetics in ARDS.[139]

Hypercapnia and the Injured Brain

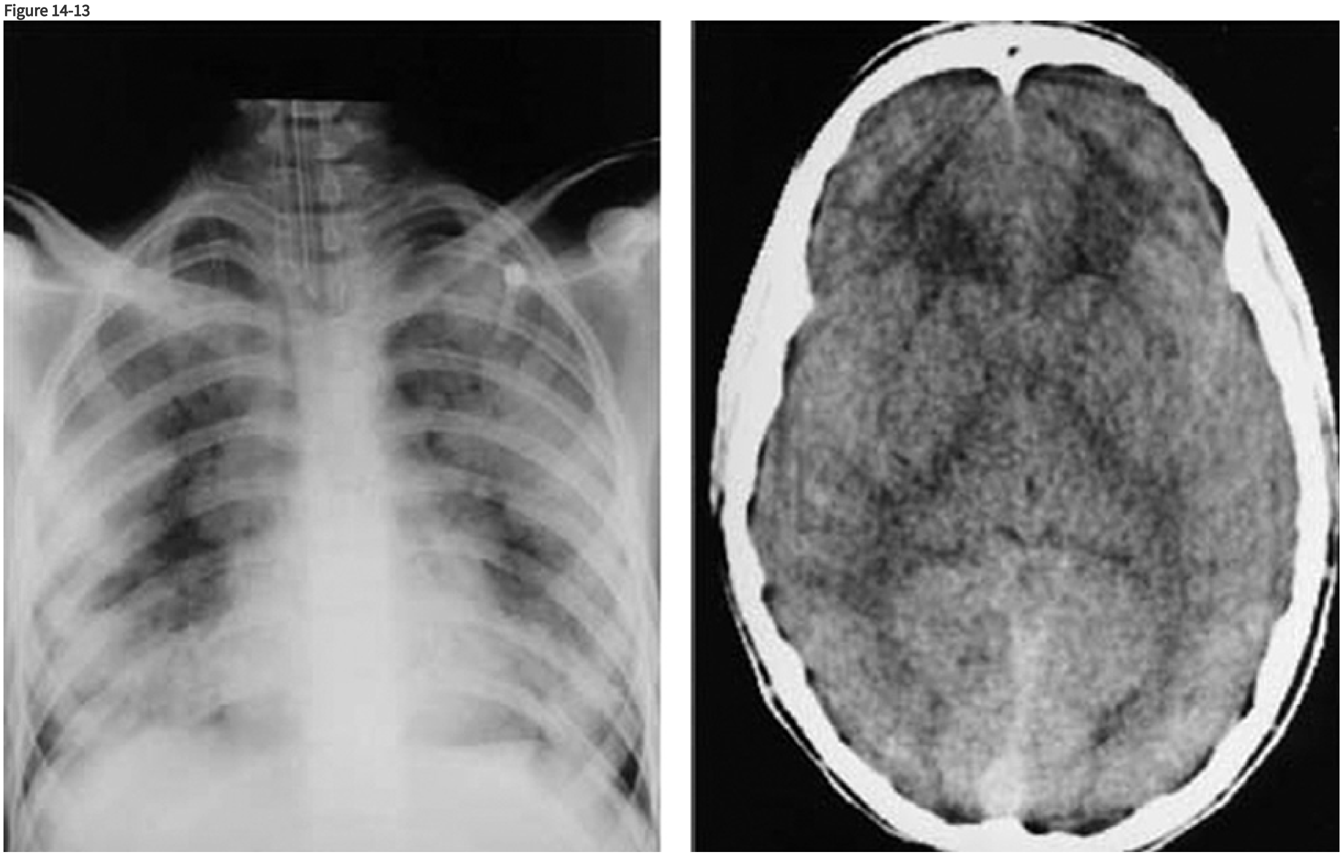
Insights from Laboratory Data

Several studies have demonstrated protective effects of hypercapnia in brain injury. Hypercapnic acidosis attenuates hypoxic-ischemic brain injury in the immature rat. Vannucci et al[142,143] developed a model of hypoxic-ischemic injury in the immature rat consisting of unilateral common carotid artery ligation, exposure to hypoxia ($F_{I_{O_2}}$ of 0.8%), and thereafter exposure to varying concentrations of inspired CO_2 (0%, 3%, 6%, or 9%) for 2 hours.[142,143] Neuropathologic assessment at 30 days of age demonstrated that hypocapnia was harmful and that elevated CO_2 was protective.[143] The experimental model caused hyperventilation. Without supplemental CO_2, the animals were frankly hypocapnic, which caused harm. Therefore, exposure to supplemental CO_2 may have provided protection by preventing hypocapnia rather than by producing hypercapnia. The investigators subsequently demonstrated that cerebral blood flow was better preserved during hypercapnia, and the higher oxygen delivery promoted cerebral glucose utilization and oxidative metabolism for optimal maintenance of tissue high-energy phosphate reserves.[142] Cerebrospinal fluid glutamate levels were also lowest with hypercapnia, and it is possible that inhibition of excitatory amino acid neuro-transmitter secretion may contribute to neural protection.[142] Additional mechanisms of neural protection may involve inhibition of free radicals[93,144] or attenuation of neuronal apoptosis.[145]

Insights from Clinical Data

Hypercapnia greatly increases cerebral blood flow and, if critical, may raise intracranial pressure, resulting in papilledema and headache. In the clinical setting, cerebral effects of hypercapnia often overlap with the effects of hypoxemia.[146] The resulting abnormalities include restlessness, tremor, slurred speech, and fluctuations of mood. Very high levels of Pa_{CO_2} cause narcosis.[146] Raised intracranial pressure, however, is not an absolute contraindication to permissive hypercapnia; in fact, with careful management it has been used successfully in a patient suffering from acute lung injury and meningoencephalitis with cerebral edema (Fig. 14-13).[147]

Figure 14-13



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

A chest radiograph and brain computed tomographic scan of a child with both ARDS and cerebral edema illustrating the trade-off that occurs when permissive hypercapnia is instituted to protect against ventilator-associated lung injury. (Reproduced, with permission, from Tasker RC, Peters MJ. Combined lung injury, meningitis and cerebral edema: how permissive can hypercapnia be? *Intensive Care Med.* 1998;24:616–619.)

Hypercapnia and the Cardiovascular System

Insights from Laboratory Data

Effects on the Myocardium

Hypercapnic acidosis protects the heart against ischemia–reperfusion injury. This has been demonstrated in isolated perfused neonatal lamb hearts, where greater degrees of hypercapnic acidosis were associated with progressively greater protection,[148] as well as in myocardial cells.[149] Although hypercapnia directly reduces contractility, it may also provide preconditioning protection.[150] Exposure to comparable metabolic acidemia (pH 6.8), however, followed by buffering to normal pH with bicarbonate and tris-hydroxymethyl aminomethane (THAM), is not protective.[148] In contrast, metabolic acidosis (pH 6.6) prevented myocardial stunning following global ischemia.[151] The latter investigators have since demonstrated that both hypercapnic and metabolic acidosis were equally effective in reducing infarct size in an in vivo canine model of coronary artery ischemia–reperfusion.[152] Possible mechanisms for the protective effects of acidosis include reduction of calcium loading to the myocardium through H^+ inhibition of calcium uptake and, in the case of hypercapnic acidosis, induction of coronary vasodilation. Nomura et al[148] found that the greatest coronary artery flow occurred with maximal hypercapnia. In contrast, increases in regional coronary artery flow do not contribute to the protective effects of normocapnic acidosis.[153]

Effects on the Vascular System

Hypercapnic acidosis attenuates the development of septic shock following cecal ligation and puncture.[24,123,125] The hemodynamic effect of hypercapnia in an ovine model was comparable to that seen with dobutamine,[108] and appears to be an effect of hypercapnia and not acidosis.[125] Hypercapnic acidosis improves microcirculation in the anesthetized rabbit,[154] which corresponds to the improvement in tissue oxygenation observed during surgery.[155,156] The absence of improved oxygenation during cardiopulmonary bypass[157] may reflect fixed aortic flow and suggest that hypercapnia-induced increases in oxygenation are a function of modulation of flow—and not vascular tone—in the perioperative setting.

Insights from Clinical Data

Critical Illness

The potential for hypercapnia to exert detrimental effects on cardiac output[158] and the peripheral circulation[159] may be overstated, particularly when hypercapnia develops gradually. As discussed earlier, the net hemodynamic effect is probably beneficial.[45,63] Nevertheless, hypercapnic acidosis may exert adverse hemodynamic effects in critically ill patients, particularly where myocardial function is already depressed. In addition, acute hypercapnia may cause hypotension where endogenous catecholamine production has been maximized or where β-blocking drugs reduce the potential for sympathetic activation to overcome the direct depressive effects of hypercapnia. Even in critically ill patients requiring inotropic drug infusions, cardiac output increases almost invariably after an acute rise in Pa_{CO_2}, although the mean arterial pressure may fall because of direct hypercapnia-induced systemic vasodilation.[136]

In patients with ARDS, the rapid induction of hypercapnia to a target Pa_{CO_2} of 80 mm Hg for 2 hours resulted in decreased systemic vascular resistance and increased cardiac output.[160] In addition, myocardial contractility was decreased, and mean pulmonary artery pressure was increased.[160] In stable patients with ARDS, reduction of V_T (from 10 to 7.7 mL/kg) associated with an increase in Pa_{CO_2} (from 37.9 to 56.7 mm Hg) was not associated with changes in hemodynamics or measures of oxygen delivery or consumption.[161]

Renal, Hepatic, and Splanchnic Effects

In isolated hepatocytes exposed to anoxia[162] or chemical hypoxia,[163] acidosis delays the onset of cell death; correction of pH accelerated cell death. This phenomenon may represent a protective adaptation against hypoxic and ischemic stress. Isolated renal cortical tubules exposed to anoxia have improved ATP levels on reoxygenation at a pH of 6.9 when compared with tubules incubated at pH 7.5.[162] Elevated

CO_2 preserves tissue oxygenation of the gastrointestinal mucosa during hemorrhagic shock[164,165] and has been reported to reverse sepsis-induced reductions in gastrointestinal cellular ATP[166] (and the reciprocal rise in lactate) and can prevent increases in mucosal permeability[167] triggered by endotoxemia.

The marked sympathetic activation associated with hypercapnic acidosis, particularly when combined with arterial hypoxemia, can lead to intense renal vasoconstriction and tubular sodium reabsorption, causing depressed glomerular filtration and increased fluid retention.[168] In a clinical study of ARDS, hypercapnia resulted in no important alterations in splanchnic circulation, although in this case CO_2 was increased by increasing apparatus dead space (and not by reduction of V_T).[169]

Effects of Acidosis versus Hypercapnia

Effects Mediated Via pH Changes

The protective effects of hypercapnic acidosis in experimental lung and systemic organ injury appear to be primarily a function of the acidosis generated.[98,112] The effects of buffering may depend on the model and the pathogenesis of the injury. In the isolated lung hypercapnic acidosis attenuated ventilator-induced lung injury, whereas in cultured cells buffering reduced cell resealing[111] and pulmonary epithelial repair[106] following physical injury. In the isolated lung, the protective effect of hypercapnic acidosis in ischemia–reperfusion was greatly attenuated when the pH was buffered,[112] and although metabolic acidosis attenuated reperfusion injury, it was less effective than hypercapnic acidosis.[112] The myocardial protective effects of hypercapnic acidosis are also seen with metabolic acidosis in ex vivo[151] and in vivo[152,153] models. Metabolic acidosis exerts protective effects in other models of organ injury; it delays cell death following caused by anoxia[162] or chemical hypoxia,[163] and reoxygenation at acidotic (compared with alkalotic) pH is associated with better preservation of cellular ATP.[162] In contrast to the lung, the type of acidosis (i.e., hypercapnic versus metabolic) appears to be of importance in the renal tubule.

Effects Mediated Via Carbon Dioxide

Certain effects of hypercapnia, including the inhibitory effects of hypercapnia on pulmonary epithelial wound healing,[106] and the deleterious effects in prolonged pulmonary sepsis,[86,87] appear to be a function of CO_2 and not pH. The effects of hypercapnia on the NF-κB pathway may also relate to CO_2 rather than acidosis,[70,106] and the developmental effects of hypercapnia, at least in *Drosophila* and in *C. elegans,* appear to be mostly independent of the pH.[70,71]

Use in Specific Clinical Settings

Acute Severe Asthma

Although much current work on ventilator strategies involving permissive hypercapnia concentrates on lung injury, its use was first described in status asthmaticus by Darioli and Pettet.[170] Permissive hypercapnia decreases dynamic hyperinflation in ventilated patients with acute severe asthma by increasing expiratory time and decreasing V_T, and this reduces end-inspiratory lung volume and lessens auto-PEEP. Other investigators have confirmed that morbidity and mortality are reduced with the use of permissive hypercapnia in ventilated patients with acute severe asthma.[171] Modest levels of permissive hypercapnia (approximately 60 mm Hg) are employed widely in ventilated patients with acute severe asthma.[81]

Acute Respiratory Distress Syndrome

The potential for protective lung ventilation strategies with varying degrees of permissive hypercapnia to improve survival in patients with acute lung injury and ARDS was suggested initially by Hickling et al.[12,13] Two studies by this group, one retrospective[12] and one prospective,[13] strongly indicated that use of low V_T was beneficial. In the retrospective study, fifty patients with severe ARDS (lung injury score ≥2.5; mean $Pa_{O_2}/F_{I_{O_2}}$ = 94) were managed with limitation of peak airway pressure to less than 30 cm H_2O.[12] The mean maximum Pa_{CO_2} was 62 mm Hg, and the hospital mortality (16%) was less than predicted by Acute Physiology and Chronic Health Evaluation (APACHE) II score (39.6%). Importantly, the authors reported no differences between survivors and nonsurvivors in terms of lung injury score, ventilator score, $Pa_{O_2}/F_{I_{O_2}}$, or maximum Pa_{CO_2}.[12] The prospective study[13] investigated comparable patients and also limited peak airway pressures to less than 30 cm H_2O, did not buffer hypercapnic acidosis, and commenced with a V_T of 7 mL/kg (as opposed to 12 mL/kg in the retrospective study). Again, hospital mortality rates were lower than predicted by APACHE II score (26.4% vs. 53.3%).[13]

The five prospective, randomized, controlled trials[10,11,172–174] that measured survival in ARDS are discussed in detail elsewhere (see Chapter 42). Two were positive (the ventilator strategy had an impact on mortality),[10,11] and three were not.[172–174] To some extent, permissive hypercapnia developed in all the trials, although there was much variability. Among the four major trials, the postrandomization values (mean ± standard deviation [SD]) in the control (higher V_T) trial groups were 35.8 ± 8.0 mm Hg,[10] 36.0 ± 1.5 mm Hg,[11] 41.0 ± 7.5 mm Hg,[172] and 46.0 ± 10 mm Hg.[174] The postrandomization values in the protective (lower V_T) trial groups were 40.0 ± 10,[10] 58.0 ± 3.0,[11] 59.5 ± 15,[172] and 54.5 ± 19[174] mm Hg. It is clear that ventilation strategy can have an impact on mortality (in the positive trials), yet there was no discernible relationship between levels of hypercapnia and survival. Also, there was no controlled approach to the administration of buffers. Finally, one of the reports mentioned (without supportive analysis) that the permissive hypercapnia might have increased the incidence of renal failure.[174]

The database of the largest of these studies[10] was subsequently analyzed to determine whether, in addition to the effect of V_T, there might be an independent effect of hypercapnic acidosis.[175] Mortality was examined as a function of permissive hypercapnia on the day of enrollment using multivariate analysis and controlling for other comorbidities and severity of lung injury. It was found that permissive hypercapnia reduced mortality in patients randomized to the higher V_T but not in those receiving lower V_T.[175] If these data are confirmed, there may be a good case for testing hypercapnic acidosis as an adjunct to attenuate ventilator-associated lung injury.

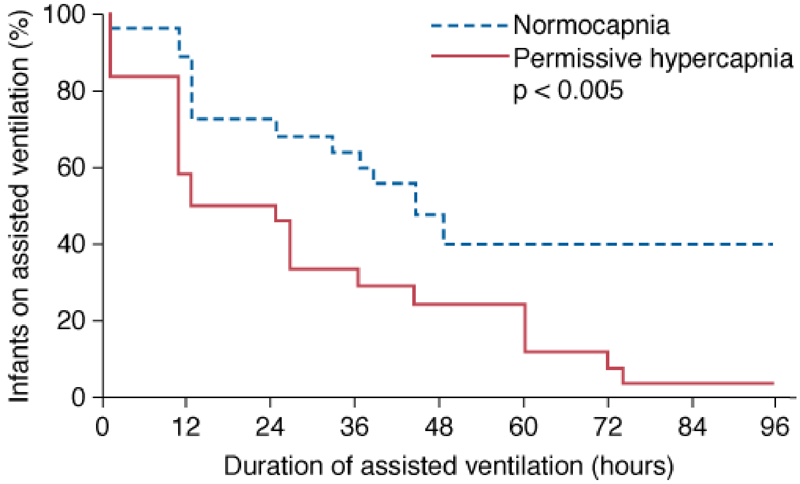
Neonatal and Pediatric Practice

Use of permissive hypercapnia in neonatology has been recognized since the study by Wung et al in 1985.[14] They described lower than previous mortality and lower incidence of chronic lung disease in neonates suffering from persistent fetal circulation who were treated with low V_T entailing a high Pa_{CO_2}.[14]

Neonatal Respiratory Distress Syndrome

Infants with neonatal respiratory distress syndrome have been studied in a randomized, controlled trial.[176] Although not powered to detect differences in survival, the duration of mechanical ventilation was shorter in the permissive hypercapnia group; and, no obvious adverse effects were seen (Fig. 14-14). Although encouraging, such a small study would not detect a subtle increase in adverse effects. More recently, a prospective multicenter study of extremely premature neonates in Denmark (1994 to 1995) reported that a ventilator strategy incorporating permissive hypercapnia and early use of nasal continuous positive airway pressure and surfactant reduced the incidence of chronic lung disease significantly.[177]

Figure 14-14



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Duration of mechanical ventilation in neonates with respiratory failure randomized to conventional therapy or permissive hypercapnia. (Reproduced, with permission, from Mariani G, Cifuentes J, Carlo WA. Randomized trial of permissive hypercapnia in preterm infants. *Pediatrics.* 1999;104(5 Pt 1):1082–1088.)

Congenital Diaphragmatic Hernia

Permissive hypercapnia plays an increasing role in the ventilator management of infants with congenital diaphragmatic hernia.[178] This contrasts sharply with traditional management, which involved aggressive hyperventilation with the aim of producing systemic alkalinization. High levels of barotrauma, poor long-term respiratory outcomes, and poor survival rates, however, have prompted the recognition that the hypoplastic lung is the major pathophysiologic defect. Accordingly, avoidance of barotrauma has assumed increasing importance, and ventilator strategies involving permissive hypercapnia are used increasingly. A retrospective analysis of three treatment protocols in high-risk infants with congenital diaphragmatic hernia reported that permissive hypercapnia was associated with a substantial increase in survival, decreased barotrauma, and decreased morbidity at 6 months. In contrast, the earlier introduction of high-frequency oscillatory ventilation (which readily controls Pa_{CO_2}) had minimal impact. Despite limitations, the increased survival associated with the use of permissive hypercapnia is persuasive.[179]

Congenital Heart Disease

Four studies of patients with congenital heart disease are relevant.[180–183] In the context of single-ventricle physiology, pulmonary vascular resistance can be controlled by inducing alveolar hypoxia or alveolar hypercapnia. Two studies documented that the addition of inspired CO_2 increased cerebral oxygenation and mean arterial pressure compared with reducing in hypoplastic left-heart syndrome[183] and following cavopulmonary connection.[182] Hypoventilation also improves systemic oxygenation after bidirectional superior cavopulmonary connection, potentially via a hypercarbia-induced decrease in cerebral vascular resistance, thus increasing cerebral, superior vena caval, and pulmonary blood flow.[180] Finally, a detailed recent study demonstrated that without altering V_T or mean airway pressure, the addition of CO_2 to inspired gas resulted in improved cerebral blood flow and systemic oxygenation following cavopulmonary connection.[181]

Complications

The complications of hypercapnia relate primarily to those caused by hypercapnic acidosis per se and those caused by the use of low V_T.

Complications Associated with Hypercapnic Acidosis

Although many physiologic effects are associated with hypercapnic acidosis, there are few complications. The most notable complications include intracranial and pulmonary hypertension. Complications are of most concern in patients with specific risk factors, especially where acidosis is acute, severe, or not buffered.

Complications Associated with Low Tidal Volumes

Reduced V_T may lead to increased intrapulmonary shunt, reducing Pa_{O_2}.[45] This is generally not a difficult problem and can be countered by a recruitment maneuver, elevation of PEEP or mean airway pressure, prone positioning, or other strategies. A more serious issue is whether small V_T values increase mortality, a source of significant controversy and a major issue in the meta-analysis by Eichacker et al.[184] They suggested that not only were very high V_T values dangerous, but so too were very small V_T values. This was the basis of a U-shaped curve depicting a relationship between V_T and mortality; the investigators suggested that the ARDS Network 6 mL/kg V_T protocol was potentially dangerous, should not be used, and should not be considered the standard of care.[184] Considerable controversy ensued. Another group published an abbreviated meta-analysis pooling the results of important clinical trials.[185] They countered the analysis of Eichacker et al and concluded that there was no statistical basis for the assertion that low V_T values are harmful.

Limitations and Contraindications

Limitations

A substantial body of literature emphasizes the potential for full recovery following exposure to extreme levels of hypercapnia, termed *supercarbia,* in both adults and children. Several children exposed to extremes of Pa_{CO_2} (155 to 269 mm Hg)[186] and one patient with asthma with a Pa_{CO_2} of 293 mm Hg (pH 6.77)[28] have been described, with excellent recovery and no long-term sequelae. In adults, in the early 1950s, the accidental development of severe respiratory acidosis was not uncommon in patients undergoing certain surgical procedures; Pa_{CO_2} as high as 200 mm Hg was reported during thoracotomy without apparent adverse effects.[187] More recently, reports exist of survival without sequelae following exposure to Pa_{CO_2} values of up to 375 mm Hg (pH 6.6).[188,189] Indeed, a case report indicates complete recovery following a pH of 6.46 secondary to ethylene glycol poisoning.[190] These numbers, impressive though they are, do not suggest that all patients—certainly not those who are already critically ill—would survive such exposure without incident or even survive at all. Nonetheless, they do indicate that severe hypercapnic acidosis per se is not *invariably* harmful.

Contraindications

As with almost any situation, there are few absolute contraindications and many relative contraindications. Some authorities suggest that intracranial hypertension is an absolute contraindication,[191] although this is disputed.[147] In any case, the risks and benefits of hypercapnia in the setting of intracranial hypertension must be weighed and monitored carefully. Additional relative contraindications include pulmonary hypertension, significant hypovolemia, or uncontrolled severe metabolic acidosis.[191]

Adjunctive Therapies

Buffering Hypercapnic Acidosis

Buffering of the acidosis induced by hypercapnia in patients with ARDS remains a common, albeit controversial, clinical practice.[192,193] The effects of buffering hypercapnic acidosis need to be considered because both hypercapnia—and acidosis—exert distinct biologic effects. As discussed above (Basic Principles), there is evidence that the protective effects of hypercapnic acidosis in ARDS are a function of the acidosis rather than the elevated CO_2 per se.[98,112] In contrast, some of the potentially harmful effects of hypercapnia, such as delayed wound healing[106] and worsening of lung injury as a result of prolonged pneumonia,[87] occur despite buffering.

Sodium Bicarbonate

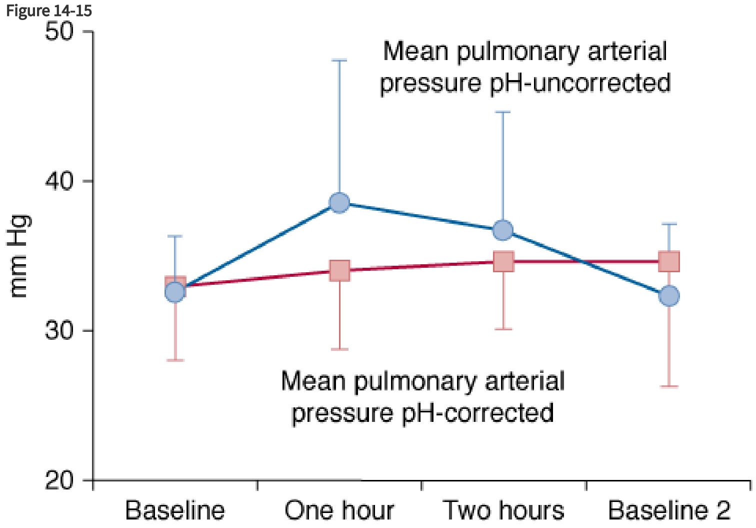
Buffering with sodium bicarbonate was permitted in the ARDS Network study.[10] Concerns with its use, however, have caused its removal from routine use in cardiac arrest algorithms.[194,195] The effectiveness of bicarbonate as a buffer depends on the ability to excrete CO_2, rendering it less effective in buffering hypercapnic acidosis. In fact, bicarbonate may further raise Pa_{CO_2} where alveolar ventilation is limited, as in ARDS.[196] Bicarbonate may correct arterial pH but worsen intracellular acidosis[197] because the CO_2 produced when bicarbonate reacts with metabolic acids diffuses readily across cell membranes, whereas bicarbonate cannot.[198]

In metabolic acidosis, the situation is also complex. Bicarbonate infusion can augment the production of lactic acid.[199–205] In ketoacidosis it can slow the clearance of ketoacids.[206] Of greater concern, bicarbonate administration is associated with a fourfold increase in risk of cerebral edema in children with diabetic ketoacidosis.[207] When compared with an equimolar dose of sodium chloride, bicarbonate administration does not improve the hemodynamic status of critically ill patients who have lactic acidosis.[208]

Tromethamine

There may be a role for the amino alcohol tromethamine (THAM) in hypercapnic acidosis. THAM penetrates cells easily and can buffer pH changes and simultaneously reduce P_{CO_2}.[209] Unlike bicarbonate, which requires an open system for CO_2 elimination in order to exert its buffering effect, THAM is effective in a closed or semiclosed system.[209] THAM rapidly restores pH and acid–base regulation in acidemia caused by CO_2 retention.[209] In a small but carefully performed clinical study of twelve patients with ARDS, rapid induction of a hypercapnic acidosis resulted in decreased systemic vascular resistance, increased cardiac output, decreased myocardial contractility, decreased mean arterial pressure, and increased mean pulmonary arterial pressure.[160] Buffering of the hypercapnic acidosis with THAM rapidly attenuated but did not fully reverse these changes (Fig. 14-15).[160] Thus, it could be argued that although permissive hypercapnia generally is well tolerated in patients with ARDS, buffering with THAM[160] might be a useful cotherapy where hemodynamic instability supervenes.

Figure 14-15



Pulmonary artery pressure is elevated by rapid institution (over 2 hours) of permissive hypercapnia (*blue line*) in patients with ARDS. Institution of a comparable degree of permissive hypercapnia buffered with tromethamine (THAM, *red line*) attenuated the effects. (Reproduced, with permission, from Weber T, Tschernich H, Sitzwohl C, Ullrich R, Germann P, Zimpfer M, Sladen RN, Huemer G. Tromethamine buffer modifies the depressant effect of permissive hypercapnia on myocardial contractility in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2000;162:1361–1365.)

Carbicarb

Carbicarb is an equimolar mixture of sodium carbonate (Na_2CO_3 0.33 M) and sodium bicarbonate ($NaHCO_3$ 0.33 M) and may have advantages over the latter. Carbicarb has been studied in mixed respiratory and metabolic acidosis and has proved effective in raising arterial pH and preventing lactate elevation;[196] in contrast to sodium bicarbonate,[197] it corrects the systemic and intracellular acidosis seen with hypercapnia[210] without elevating Pa_{CO_2}. Carbicarb, however, did not appear to possess any advantages in terms of restoration of hemodynamic stability over isoosmolar amounts of hypertonic saline or sodium bicarbonate in a canine shock model.[201] The similar responses may relate to their identical sodium content, with no additional benefit attributable to correction of pH.[201]

In summary, although common clinical practice, no outcome data (e.g., survival or duration of hospital stay) support the buffering of hypercapnic acidosis in patients. In the absence of correcting the primary problem, buffering hypercapnic acidosis with bicarbonate is not likely to be of benefit. If the clinician elects to buffer a hypercapnic acidosis, the rationale should be clear (e.g., to ameliorate potentially deleterious hemodynamic consequences) and the responses measured. THAM and Carbicarb may have a role in these clinical situations.

Augmenting Carbon Dioxide Clearance

Dead Space Gas Replacement

At the end of expiration, the ventilator circuit distal to the Y-piece and the anatomic dead space both contain alveolar gas. This CO_2-rich gas then constitutes the first part of the next breath delivered to the distal lung. The contribution of this dead space gas to ventilation increases with decreased V_T, given that dead space is relatively fixed. Techniques that aim to replace dead space gas with fresh gas have been advocated as an adjunct to protective ventilator strategies. These techniques may increase effective alveolar ventilation and facilitate further reductions in V_T, minimizing transpulmonary pressures.

Tracheal gas insufflation (TGI) delivers fresh gas into the central airways either continuously or in a phasic fashion during expiration (see Chapter 22). Experimental studies in animal models of ARDS and in lung models highlight the potential role of TGI in clinical practice.[211] Zhu et al[99] reported that TGI, either alone or in combination with partial liquid ventilation, attenuated the development of lung injury resulting from mechanical ventilation of surfactant-depleted lungs. Use of expiratory TGI can help clear CO_2 and facilitate V_T (and plateau pressure) reduction in patients with combined brain and lung injury, and it has also been demonstrated to augment CO_2 clearance when combined with high frequency oscillation. Despite extensive investigation, however, concerns persist with regard to the safety and monitoring of TGI that have impeded its introduction into clinical practice.[212]

Aspiration of dead space gas during expiration and controlled replacement with fresh gas is a related technique designed to minimize dead space. A feasibility study of eight patients with chronic obstructive pulmonary disease who were managed with permissive hypercapnia demonstrated that aspiration of dead space gas resulted in a similar decrease in Pa_{CO_2} but with less intrinsic PEEP compared to TGI.[213]

Coaxial double-lumen endotracheal tubes, which eliminate the contribution to dead space from the ventilator circuit distal to the Y-piece, may improve the efficiency of ventilation. No adverse hemodynamic effects or auto-PEEP has been detected with use of the coaxial tube.[214] Reduction in Pa_{CO_2} is inversely proportional to V_T. Initial safety and efficacy evaluations of this promising adjunct in patients with ARDS are required.

Additional Techniques

High-frequency oscillatory ventilation and extracorporeal CO_2 removal are discussed in Chapters 19 and 21, respectively.

Adjustments at the Bedside

Adjustments at the bedside are specific to the patient and the clinical context. The following principles may guide therapy, although in the end the physician must individualize the risks and benefits for each patient.

First, in a patient who is being mechanically ventilated, confirm correct placement of the endotracheal tube and the ventilation of both lungs. Consider and (where possible) correct reversible conditions that might have an impact on oxygenation or CO_2 removal (e.g., pleural effusion, atelectasis, and/or pneumothorax). Next, decide whether the patient is comfortable and whether there is patient–ventilator dyssynchrony. Then note the ventilator settings (i.e., plateau pressure, V_T, and rate) and the arterial blood-gas values, and consider whether these are appropriate. Concurrent with these considerations, evaluate the patient for causes of high, and reduce or eliminate any that are present. Such conditions include increased production of CO_2 (e.g., fever, sepsis, shivering, or more rarely, thyroid storm, malignant hyperthermia, or neuroleptic malignant syndrome) or decreased CO_2 elimination (e.g., increased dead-space-to-tidal-volume ratio [V_D/V_T]).

The next step involves arbitrary decisions made by the clinician at the bedside on the target tidal distension (V_T, or plateau pressure). It is important to recognize that optimal values are unknown, which is not to say that the values are unimportant; they are definitely important. Across populations, adverse effects are associated with very high V_T.[10,11] We do not advise on the ideal V_T or plateau pressure for any specific patient, recognizing that these issues can be both difficult and controversial. After clinicians select an appropriate V_T for the patient in question, they next consider the "maximal" allowable Pa_{CO_2} and the degree of acidosis that they believe the patient can tolerate. This is empirical. Although some authors suggest arbitrary limits, in the end, the decision is based on patient comfort, presence of contraindications, and clinician "comfort." When considering ventilator settings, consider the relative values of V_T, plateau pressure, and rate, and estimate the relative contributions of changes in each. For example, if the rate is inappropriately low, then increasing it will allow further reduction in V_T with less elevation in Pa_{CO_2}. Conversely, increasing the rate may induce a degree of hyperinflation (auto-PEEP),[215] which, although potentially protective, can complicate ventilator management and could induce elevation of the Pa_{CO_2}.

Finally, having decided on the desired ventilator settings and the permissible limits of toleration, decide on the trade-offs involved and the time scale. One trades the benefits of the approach—reduction of ventilator-associated lung injury—against the cost—conditions caused or exacerbated by permissive hypercapnia. Such trade-offs are inherent in the practice of medicine[216] but are seldom addressed by clinical trials. Finally, the clinician decides on the rate of introducing permissive hypercapnia. In some situations, there is little choice. For example, in acute severe asthma, hypercapnia occurs immediately, for otherwise, the patient would be subjected to incredibly high airway pressures. Having decided on the target V_T or plateau pressure, increase the rate and slowly decrease the V_T. There is no formula. Extremes of V_T or plateau pressure should be reduced immediately, and less extreme elevations should be reduced more gradually. For example, in ARDS, a V_T of 15 mL/kg could be reduced immediately to 10 mL/kg and further reduced over several hours with a concomitant increase in rate. Obviously, if ideal ventilator settings are tolerated by the patient and the and pH are normal, then the clinician would not do anything to elevate the Pa_{CO_2}.

Troubleshooting

To address problems that develop, it is key to understand the pathophysiology. This amounts to management of dyspnea, intracranial hypertension, pulmonary hypertension, and sometimes, hypoxemia. Troubleshooting follows logically the principles used in initiating permissive hypercapnia. In reality, it represents a work in progress, wherein the clinician considers risks versus benefits (real and potential) for each patient at the bedside.

Tidal Volume and Respiratory Rate

The first issue is to determine whether the V_T (or perhaps plateau pressure) is actually in the range desired by the clinician. If V_T is far lower than desired, increasing it will alleviate "unnecessary" hypercapnic acidosis. If the respiratory rate is too low, increasing it also will lower the Pa_{CO_2}.

Rate of Change

Was the permissive hypercapnia introduced too quickly? Overly rapid introduction of hypercapnia results in a greater degree of acidosis because the physiologic buffering systems are unable to cope. Most unfavorable effects of hypercapnic acidosis (e.g., dyspnea and intracranial hypertension) are more pronounced where acidosis is greater. More gradual introduction of permissive hypercapnia may prevent these problems.

Adjuvant Therapies: Control of Metabolic Acidosis

Buffering sometimes can be appropriate for modifying the adverse effects of significant acidosis. The clinician needs to identify conditions other than ventilator settings and Pa_{CO_2} per se that have an impact on acid–base status. Hyperchloremic acidosis may have developed consequent to high volumes of intravenous saline, total parental nutrition, or renal tubular acidosis, and can be buffered easily or possibly prevented. Renal impairment, which slows the generation of endogenous bicarbonate or excretion of hydrogen ion, or generates endogenous organic acids, can be alleviated by renal replacement therapy.

Adjuvant Therapies: Alternative Elimination of Carbon Dioxide

Adjuvant therapies can be directed at the elevated Pa_{CO_2}. Examples include high-frequency oscillation, TGI, and extracorporeal techniques.

Specific Evaluation of the Complication

In some circumstances, clinicians suspect a problem, such as pulmonary or intracranial hypertension, that leads them to completely avoid permissive hypercapnia. Such an approach may not be appropriate in the absence of clear proof of the feared condition because the patient may be at a far greater risk of ventilator-associated lung injury than of the feared condition or potential complication. In such a setting, the clinician should consider appropriate monitoring. For example, insertion of an intracranial pressure monitor or jugular oximetry catheter may provide evidence that allows the clinician to gradually introduce, titrate, or clearly avoid hypercapnia in a patient with head injury. Such an approach has been described when managing children suffering from meningococcal septicemia complicated by elevated intracranial pressure and severe acute lung injury (see Fig. 14-13).[147] Concerns about pulmonary hypertension can be addressed by measuring, and monitoring the degree of pulmonary hypertension or its sequelae (e.g., right-ventricular failure, tricuspid regurgitation, or right-to-left shunting). In this case, transthoracic echocardiography or pulmonary artery catheterization may be indicated. The clinician will recognize that adverse effects (e.g., epithelial fluid reabsorption) may exist, which may not be detected by conventional monitoring.

Specific Treatment of the Complication

Direct testing for a feared complication may enable early detection. It also permits the direct “independent” treatment if permissive hypercapnia is still deemed necessary to tolerate specific ventilator settings. Specific treatment might include inhaled nitric oxide for pulmonary hypertension[136] or sedation, osmotherapy, or hypothermia for intracranial hypertension. Of course, the most common complaint, dyspnea, can be treated directly with sedatives and opioids[217]; neuromuscular blockade, while paralyzing respiratory muscle contraction, of course, does not alter the perception of dyspnea.

Important Unknowns

Although much is known about hypercapnia, much remains unknown. The principles underlying the approach to permissive hypercapnia have been established.[25,31,33,191,218–220] Such principles are inherently limited by the state of our knowledge. The unknowns relating to permissive hypercapnia can be grouped as follows.

Hypercapnia versus Acidosis

Most of the acute physiologic effects of hypercapnic acidosis can be attributed to the effects of pH. CO_2 itself, however, has significant biochemical effects, especially on tissue nitration. In the context of endogenous or exogenous buffering, the effects of elevated CO_2 are not understood.

Therapeutic Hypercapnia

The role of the deliberate elevation of CO_2, as opposed to passive elevation of CO_2 resulting from reduced V_T, is experimentally exciting but essentially conjectural.[25] The state of knowledge is not sufficient to justify deliberating elevating CO_2 other than as a means of avoiding ventilator-associated lung injury; clearly, in the context of evolving understanding of the relevant mechanisms, clinical study with mortality as an outcome would not be justified.[135]

Mechanisms of Benefit (or Harm)

Although evolving studies are providing better understanding of the mechanisms of benefit and harm associated with hypercapnia and acidosis, clinical definitions of lung injury do not take into account the biologic processes that cause the injury. Although some mechanisms are understood, it is not yet possible for the clinician to extrapolate from mechanisms discovered in the laboratory to the care of a patient with lung injury.

Monitoring during Hypercapnia

There are no specific monitors for patients being ventilated with permissive hypercapnia. Regular arterial blood-gas analysis is a requirement, and clinicians will monitor parameters that lead them to detect or prevent complications such as intracranial hypertension.

The Future

Future advances in the early diagnosis of acute respiratory failure and the development of specific therapies may reduce the need for ventilator strategies involving permissive hypercapnia. In the interim, ventilator strategies involving hypercapnia appear likely to play a central role in the management of these disease states. This highlights the need to improve our understanding of the biology of hypercapnia, which should lead us to a better understanding of the advantages, disadvantages, and contraindications pertaining to its use in the clinical context. If the requirement for adverse ventilator strategies lessens, the requirement for permissive hypercapnia will lessen in parallel, and concerns related to its use will become less relevant. Nonetheless, a fuller profile of biochemical and physiologic responses to elevated CO_2 will be needed for the clinician of the future to decide whether hypercapnia is dangerous—or potentially beneficial—in the management of a specific patient.

Summary and Conclusions

Permissive hypercapnia simply means reducing V_T in order to lessen the likelihood of ventilator-associated lung injury; in so doing, the clinician accepts the inevitable development of higher Pa_{CO_2}. Elevated P_{CO_2} is associated with a long list of physiologic alterations. Some of these effects are harmful, others may be neutral, and some may turn out to be beneficial. In any case, there is ample evidence that high tidal volumes harm patients. In avoiding such high tidal volumes, the clinician must decide for each patient what is the appropriate trade-off between the benefits of avoiding high V_T and the cost (and benefits) of the associated hypercapnia.

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